



Dirty air and green investments: The impact of pollution information on portfolio allocations[☆]

Raymond Fisman^a, Pulak Ghosh^b, Arkodipta Sarkar^{c,*}, Jian Zhang^d

^a Department of Economics, Boston University, 270 Bay State Road, Boston, MA 02215, United States

^b Indian Institute of Management Bangalore, Bannerghatta Road, Bangalore, 560076, India

^c NUS Business School, National University of Singapore, 15 Kent Ridge Drive, Singapore, 119245, Singapore

^d Faculty of Business and Economics, The University of Hong Kong, Pokfulam Road, Hong Kong, China

ARTICLE INFO

Dataset link: <https://data.mendeley.com/datasets/kwjwcn3wgy/2>

JEL classification:

D83

G11

Q53

Keywords:

ESG

Sustainable investing

Pollution information

Saliency

ABSTRACT

We study whether access to local pollution information causes investors to make greener portfolio allocations, exploiting the rollout of air quality monitoring stations in India. Using a triple-differences framework on the trading records of 19 million investors, we show that retail investors' holdings in "brown" stocks become more negatively related to local pollution after a nearby station appears. This effect is more pronounced on "alert" dates when air quality is reported to be harmful. The effect is strongest among tech-savvy investors likely "treated" by real-time pollution data, and younger investors, who may be more sensitive to environmental concerns.

1. Introduction

Interest in so-called ESG (environmental, social, governance) investment has grown immensely in recent years. One oft-cited indication of this growth is that assets under management at ESG funds have increased by over 500 percent, from US\$4.8 trillion in 2010 to US\$29.2 trillion in 2021, nearly three times the growth rate of assets under management more generally.¹

There is an active debate, however, on the extent to which this growth involves an actual change in investor allocations or simply reflects marketing and/or "greenwashing" efforts (Krueger et al., 2020; Bolton and Kacperczyk, 2021; Starks et al., 2023; Pastor et al., 2023).² This debate has drawn particular attention among academic scholars and policymakers to the incentives, costs, and benefits associated with

sustainable finance (Goldstein et al., 2022; Pástor et al., 2021; Pedersen et al., 2021; Giglio et al., 2023; An et al., 2024). Ultimately, this discussion requires a better understanding of the underlying household demand for green behavior by firms and other financial market participants, as this influences fund behavior through the preferences of its clientele and also has a direct impact through retail investor allocations.³ Crucially, recent evidence suggests that retail investors may counteract the shift toward green investment by institutions, potentially undermining progress toward sustainability goals (Pastor et al., 2023). To the extent that asset prices and/or firms' real activities are affected by these shifting investment patterns, it is an important policy matter to better understand whether particular interventions may encourage increased ESG orientation for individual investors. Such policies

[☆] We are especially grateful to an anonymous referee and the editor, Dimitris Papanikolaou for many thoughtful comments and suggestions. We also thank the comments from Reena Aggarwal, Sumit Agarwal, Meghana Ayyagari, Maxime Bonelli, Laurent Calvet, Darwin Choi, Dimas Fazio, Sangeun Ha, Chanik Jo, Tse-Chun Lin, Poorya Kabir, Abhiroop Mukherjee, Larry Qiu, Rodimiro Rodrigo, Zacharias Sautner, Christoph Schiller, Ariel Weinberger, Vikrant Vig.

* Corresponding author.

E-mail addresses: rfisman@bu.edu (R. Fisman), pulak.ghosh@iimb.ac.in (P. Ghosh), asarkar@nus.edu.sg (A. Sarkar), zhangj1@hku.hk (J. Zhang).

¹ See <https://www.unpri.org/about-us/about-the-pri>, accessed April 23, 2025.

² Pastor et al. (2023), for example, suggest that there is only a 6 percent "ESG tilt" to assets under management, driven by the largest institutional investors.

³ See, for example, the discussion of Vanguard's decision to withdraw from the Net Zero Asset Management Initiative: <https://www.reuters.com/business/sustainable-business/vanguards-climate-group-exit-shows-retail-investors-trail-esg-2023-01-12/>, accessed April 23, 2025.

could align retail investor behavior with broader sustainability objectives, ensuring that household investments complement rather than offset institutional shifts toward greener portfolios.⁴

In this paper, we contribute to our understanding of what drives individuals' ESG investment and also provide insight into the factors that encourage it. To do so, we examine the role of pollution salience in investors' "green" allocation decisions, via a setting that affords a clear econometric identification as well as policy relevance. Specifically, we study how the advent of a technology that provided real-time pollution information that plausibly made environmental concerns more salient affected investment in "green" versus "brown" companies.

The arrival of real-time pollution information to investors came via the introduction of continuous ambient air quality monitoring stations (CAAQMSs) across India. The first monitoring station appeared as a pilot program in 2007, with the full rollout taking place during the 2010s. We posit that the arrival of a monitoring station makes pollution more salient for those located nearby, who receive easy access to air quality data. Crucially, given our focus on the link from information salience to investor behavior, there was no change in actual pollution around a monitoring station's arrival. We argue that, whereas pollution may have simply been ignored prior to the appearance of CAAQMSs, environmental concerns were made more salient as a result of easy-to-access real-time air quality information (e.g., via smartphone or billboard). These readily available pollution updates, in turn, may lead investors to take a greater interest in "greener" investments. Helpful for identification, the monitoring stations' readings were advertised as relevant for a range of 20 kilometers, offering a natural way of defining investors who are "treated" with access to pollution information, which we compare to the portfolio allocations of "control" investors that are more distant from monitoring stations (but still close enough to the treatment population that they plausibly serve as a relevant benchmark – we also show that our results are robust to using a tighter range for our treatment definition).

We utilize the complete set of trading records from the National Stock Exchange of India (NSE) to construct the portfolios of approximately 19 million investors during the years 2004–2020. We can trace any trades in these portfolios to an individual's (anonymized) permanent account number, and the associated Postal Index Number (PIN) Code for the account holder. We can thus calculate, with a high degree of precision, the distance between an investor's address and the nearest CAAQMS, and evaluate how the sensitivity of their portfolio to air quality changed after the monitoring program made this information widely available. As we explain below, we observe — albeit imperfectly — air quality prior to the creation of CAAQMSs via satellite data, though importantly, data from these stations were not made available to the public in real-time. This allows us to distinguish the role of salience from shifts in actual pollution that might be correlated with the appearance of monitoring stations.

Our main focus is to evaluate how access to pollution information affects the extent to which household ESG investments respond to short-term pollution fluctuations, i.e., we are interested in the brown-share-pollution gradient $\frac{\partial \text{BrownShare}}{\partial \text{Pollution}}$. This measurement choice of portfolio share aligns with the theoretical ESG-augmented framework established in earlier work, in which investors derive a warm-glow utility from holding green stocks, making the weighted sum of ESG scores for all equity holdings a critical factor for portfolio allocation (e.g., [Pedersen et al., 2021](#)).⁵ Specifically, we examine the sensitivity of an

investors' holdings in "brown" stocks to air pollution severity before versus after the opening of a nearby monitoring station. In an event study specification, we show that the brown share of investors in an impacted PIN Code is unrelated to air quality in the two years leading up to the appearance of a station. With the station's appearance, the air quality-brown share relationship immediately becomes negative, and remains so throughout the rest of the sample period. Our point estimates indicate that, for a one standard deviation decline in (within-monitoring-station) air quality, there is a 0.5 percentage point (1.25 percent of the mean) decrease in the share of brown stocks in affected investors' portfolios after a monitoring station appears (whereas the sensitivity is approximately zero beforehand). We use a "donut hole" approach to define a benchmark set of investors, located in PIN Codes in the 40–60 kilometer range around a given monitoring station—these investors are sufficiently distant from the station that its reading had less relevance (as noted above, the range communicated to the public was just 20 kilometers). For this group of "donut hole" investors we observe no change in investment sensitivity to air quality after a monitoring station opens. Finally, we show that these net portfolio shifts are driven entirely by brown stock sales rather than green stock purchases. Since green stocks do not outperform brown stocks over this period, we suggest that our findings are most plausibly driven by investor tastes and pollution salience rather than a shift in expected returns.

To link these patterns more directly to pollution salience, we examine shifts in portfolio allocations around changes in particularly salient shifts in air quality. Specifically, we look at air quality "alert" days when the air quality reading transitions from Moderate (yellow) to Poor (amber), a change that also triggers government warnings about the health risks posed to the general population. Using a regression discontinuity approach around this air quality cutoff, we show that there is a discrete drop in the share of brown stocks held by retail investors.

In a final set of results, we explore heterogeneity in investors' responses to exposure to air quality information. Given that air quality information from monitoring stations was delivered primarily via a smartphone app, we suggest that salience may have been greater among tech-savvy investors who were also more apt to trade via mobile phone. Thus, we begin by splitting the sample based on whether a trader most often executed transactions via mobile device, the internet, or by some other means (i.e., through either a trading kiosk or a broker). We find a far greater shift in responsiveness to pollution after a monitoring station appears amongst mobile-based investors; there is also a greater sensitivity for internet-based investors relative to those using more traditional methods.

We next explore heterogeneity by age. Beyond a moderately higher degree of tech savvy, young investors may also have greater environmental awareness since they disproportionately bear the consequences of climate change and environmental degradation. We thus posit that the young will be more responsive to being confronted with pollution information.⁶ We find that air quality information has the greatest effect on investment sensitivity for young investors, and the weakest effect for the elderly. Finally, we split the sample by gender. While the motivation for this heterogeneity test is less straightforward than our other sample splits, we suggest that it links to the broader literature on the "environmental gender gap"—women express greater concern for environmental issues than men (e.g., [Xiao and McCright, 2015](#)). We extend this line of research to show that this gender gap applies to male versus female investors: the portfolios of women are more sensitive to air quality than those of men, once pollution information becomes readily available.

⁴ See, for example, [Akey and Appel \(2019\)](#), [Naaraayanan et al. \(2021\)](#), [Heath et al. \(2023\)](#), [Kahn et al. \(2023\)](#), [Hartzmark and Shue \(2023\)](#), [Kojien et al. \(2023\)](#), [Oehmke and Opp \(2024\)](#).

⁵ Alternatively, this measurement choice can also be motivated by a simple discrete choice model of green versus brown stocks, which generates the pooled average of choices at the household level, i.e., the brown share ([Berry, 1994](#)).

⁶ Alternatively, air quality may be more salient for the elderly, given the health consequences, though to the extent that this is the case, our results suggest that it is dominated by the aforementioned effects.

Our work bridges two literatures—the body of research that aims to understand investors’ non-pecuniary concerns generally, and specifically their interest in ESG investments, and work that explores the role of information salience in judgment and decision-making. Focusing on the determinants of ESG investment, our research is most directly related to work linking personal preferences to portfolio choices. Several previous studies in particular have shown how personal circumstances and experiences impact portfolio allocation. Andersen et al. (2024), for example, show that having offspring with respiratory disease leads to a decrease in parents’ ownership of brown stocks, while Choi et al. (2020) find that higher temperatures are associated with a decline in prices of “brown” stocks, driven by the trading activity of retail investors. These findings, based on plausibly exogenous shocks to individual circumstances (whether idiosyncratic or more aggregate), are complemented by field and survey evidence which suggest that these shocks influence allocations via preferences rather than return expectations (Bauer et al., 2021; Giglio et al., 2023). We differ from this earlier work in that we highlight the crucial role of information dissemination in mediating the link from environmental (or other) conditions to investment behavior. That is, in a sense the analysis of Choi et al. (2020) is analogous to the pollution-investment correlation during the pre-treatment baseline period in our study, so that we show the importance of making information on environmental problems more readily available for retail investors’ trading decisions. Given that we find no investor response to pollution during this baseline period, our results suggest a revisiting of the question of whether investors will indeed respond at all to environmental conditions in the absence of a salience intervention. Our findings also have more direct policy implications, since highlighting social concerns is a policy variable, rather than an exogenous set of circumstances, like sickness or temperature. Our individual-level data further allow us to explore the types of investors that are most responsive to making pollution data available. The data are themselves an important distinguishing feature of our work. Because we have access to high-frequency trading data for the universe of retail investors on one of the largest exchanges in the world, we may examine how ESG-relevant information impacts portfolio choice with a granularity that was not possible in prior work.

We contribute to two areas within the broader literature on salience and decision-making: how information affects investor decisions, and the link from information to environmental concerns more broadly. While our focus is on pollution salience, there is a much larger literature that examines how inattention and salience impact portfolio allocation decisions, whether driven by the notability of past returns (Bordalo et al., 2013; Cosemans and Frehen, 2021); the media (Huberman and Regev, 2001; Kang et al., 2024); or ESG ratings themselves (Hartzmark and Sussman, 2019; Pelizzon et al., 2021). The last set of results suggest other potential approaches to encouraging the incorporation of environmental concerns in investment decisions. Hartzmark and Sussman (2019), for example, finds that when Morningstar first began publishing its sustainability ratings, funds receiving a “5-globe” rating received large inflows and those receiving 1-globe ratings suffered symmetrically large outflows. Policies that impose the posting of ESG information by data providers may thus be a complementary salience-based policy to guide investors on how to act on increased environmental concerns.⁷

Beyond finance-focused applications, our work contributes to the larger literature that explores whether exposure to environmental shocks impacts beliefs and attitudes towards climate change (Li et al., 2011; Zaval et al., 2014; Lujala et al., 2015). Going from attitudes to actions, Chang et al. (2018) examine how daily variation in air

pollution impacts the demand for health insurance, and Barwick et al. (2024) examine how pollution information — delivered by the same type of real-time monitoring stations that we consider here — impacted online search behavior in China, with more searches related to pollution avoidance behaviors after a monitoring station appears. Our work also documents real behavioral changes, but in a very different domain, and one that has broader social consequences rather than one with private benefits. In line with these earlier works, our findings suggest that policies that give greater visibility to environmental quality may be useful in encouraging “greener” behaviors. While our setting involves investment, it is plausible that one may observe similar shifts in, say, energy use or other consumer behaviors.

2. Institutional background

This section provides an overview of India’s rollout of continuous ambient air quality monitoring stations (CAAQMSs), which we exploit in our empirical design. The introduction of CAAQMSs was part of the broader National Air Quality Monitoring Programme (NAMP) set up by the Central Pollution Control Bureau (CPCB) in coordination with state-level control boards. As we discuss in this section, the initiative led to a rapid and significant increase in the availability of pollution data to the general public, resulting in a considerable improvement in awareness of pollution problems.

A Brief History of Air Quality Monitoring in India: The CPCB has been systematically collecting pollution data under its national monitoring program since 1985.⁸ Initially, monitoring stations collected data on four key pollutants — sulfur dioxide (SO₂), nitrogen dioxide (NO₂), PM₁₀ (particulate matter under 10 microns), and PM_{2.5} — via manual monitoring stations scattered throughout the country. Generating air quality estimates via these manual stations involved the collection of ambient air over a period of several days, which was then transported to a central location where the data were analyzed. The resultant (manually generated) report was then archived in the Environment Air Quality Data Entry System. This data collection process took weeks to complete, so that much of the information was stale by the time the report was finished. Furthermore, the reports themselves were often inaccessible—it was left to local pollution control authorities to upload and/or make them available. Finally, the data were, in some cases, constrained or unsuitable for analysis, so no data were available at all (Pant et al., 2019). The CPCB itself released the data at a monthly or annual frequency.

As the preceding description makes clear, the manually collected pollution data gathered under the earlier NAMP were slow to produce, of questionable quality, and hard for the general public to access (if the data were made available at all).

Both the quality and availability of data shifted markedly with the introduction of CAAQMSs, first piloted in 2007 and later expanded in 2016 (see below for more details on the rollout). These newer monitoring stations collect information on a wider range of pollutants than earlier instruments.⁹ Furthermore, both the collection and analysis of data were fully automated via “internet of things” technology that facilitated continuous automated data collection, as well as the transfer of pollution data to a central server in real time every few minutes. At the center, data analysis was also automated, and ready for use shortly after collection. Pollution data from the CAAQMSs are used to calculate the air quality index (AQI), a standardized metric that incorporates a wider range of pollutants than the earlier NAMP monitoring program. Real-time AQI readings are publicly available online and via smartphone

⁷ The question of whether and how retail investors generally respond to ESG information in the absence of the sort of very strong salience provided by, e.g., their prominent presentation on Morningstar listings, remains a topic of active research. See, in particular, Moss et al. (2023) and Li et al. (2023).

⁸ Air quality monitoring began earlier, in 1978, in 10 cities—Ahmedabad, Mumbai, Kolkata, Kochi, Delhi, Hyderabad, Jaipur, Kanpur, Chennai, and Nagpur.

⁹ Measured pollutants including PM_{2.5}, PM₁₀, nitrogen oxides, sulphur dioxide, benzene, toluene, ethylbenzene, and xylene.

apps, with historical data archived by the CPCB and downloadable on its website.¹⁰

A primary purpose of the continuous air quality monitoring program is to create public awareness of environmental conditions. This objective was greatly facilitated with the advent of CAAQMSs, as the earlier manually-driven system could not provide real-time environmental data to the public. This information is now delivered via public displays, web widgets, alerts, and a proliferation of mobile apps that provide localized information on air quality.

Rollout and location of CAAQMSs: Real-time pollution monitoring was first piloted in 2007, with the expansion accelerating in 2016 (see Fig. 1).¹¹

The decision of where to locate monitoring stations is made by the CPCB in consultation with the state-level pollution control boards. The criteria naturally include consideration of nearby pollution severity and population; however, there is a much longer list of practical concerns that include geographic obstructions, security, cost, and power availability, among many others. While we identify our main results based on a triple-differences framework, it is still worth noting that within a relatively narrow geographical region, the locations of CAAQMSs (as well as their timing) are dictated in large part by considerations that are largely exogenous to factors that might affect the portfolio allocation decisions of individual traders.¹²

Classification of Treated versus Control PIN Codes: Our empirical approach is to compare individuals located in the immediate vicinity of newly established CAAQMSs (i.e., the treated group) to individuals who are marginally further away, and thus “untreated” with pollution information (i.e., the control group). There is no clear threshold for how the relevance of air quality readings diminishes with distance from a monitoring station. We propose a donut-shaped approach that leverages the comparison of average pollution levels within the inner and outer rings of the ‘donut,’ while leaving out the area in the middle as the buffer region. We use a 20-kilometer radius of a CAAQMS to define traders who are “treated” with AQI information when a monitoring station opens. This range was determined based on discussions with CPCB officials, who suggested that this is the distance that pollution readings would be perceived as relevant to the public. We use investors located 40 to 60 kilometers from a monitoring station as the benchmark or control group.¹³

¹⁰ It is difficult to measure exactly how many people keep track of local air quality via smartphone because, in addition to the CPCB’s own app, there are dozens more that provide real-time AQI data for India and internationally. Many are listed as having 100,000+ downloads. The CPCB reports that its own app has been downloaded over 500,000 times. For more details on the monitoring program as well as references for AQI measurement, see Pant et al. (2019).

¹¹ See, e.g., Gulia et al. (2022) on the timing of CAAQMSs. For the current map of stations, see <https://airquality.cpcb.gov.in/ccr/#/caaqm-dash-board-all/caaqm-landing>, last accessed April 23, 2025.

¹² Greenstone et al. (2022) similarly use the rollout of monitoring stations in China to study the impact of real-time information on local underreporting of pollution, and also online searches for, e.g., face masks and filters.

¹³ These cutoffs are not definitive, and for this reason we offer a number of robustness checks to show that our results are not sensitive to these particular choices. We choose our main cutoffs based on our reading both of Indian government communication as well as prior research on a monitoring station’s relevant radius. For example, in a UNICEF report that examines the proportion of children under threat to respiratory infections from pollution, Rees et al. (2019) highlight the lack of reliable data on ground-level monitoring of pollution through air monitoring stations. In this context, the report uses 10 kilometers as the first relevant radius around the monitoring station and 50 km as the upper limit. In the context of India, Roychowdhury et al. (2023) argue that a CAAQMS has a range of 10 kilometers, and if there are no topographical barriers the range could be up to 50 kilometers. Further, in most mobile apps the name of the monitoring station is mentioned, which makes the listed AQI feel less relevant for a resident situated a significant distance away. We take

There are cases of overlapping treatment and control areas in some densely populated regions, as India’s major metropolitan areas are covered by multiple stations. For example, Delhi had 41 monitoring stations to cover its 20 million residents spread across nearly 1500 square kilometers. There were 21 stations for Mumbai (population 21 million, area 603 square kilometers); and 7 stations for Kolkata (population 15 million, area 205 sq km). We describe below how we deal with the handful of PIN Codes where this issue arises.

Note, however, that once one gets outside of a small number of major metropolitan areas, coverage is much sparser. For example, Jodhpur, a city of 1.6 million population, had a single monitoring station, and there were no other stations within 60 kilometers of it. The entire state of Jammu and Kashmir had just a single monitoring station, in its largest city of Srinagar (population 1.7 million).

Before proceeding to the next section, we emphasize that our results do not depend on this specific treatment-control definition. We also consider narrower radii of 5, 10, or 15 kilometers in robustness checks below (Appendix Table A8). Furthermore, since India’s largest cities have a preponderance of overlapping monitoring stations, we demonstrate that our results still hold when we exclude traders in the country’s 10 largest cities.

3. Data

We use several data sources in our analysis that allow us to connect pollution, monitoring, and trading behavior at a granular level. The databases include individual-level stock holdings over time for Indian investors as well as basic geographic and demographic information; geocoded information on the timing of the installation of CAAQMSs, and information on local pollution inferred from satellite images.

Stock Trading Data: We use comprehensive data on stock trading from the National Stock Exchange (NSE) of India, one of the largest stock exchanges in the world.¹⁴ We obtained data on the universe of trading records from the NSE for the period of 2004–2020. The data allow us to observe a number of features for each transaction—the anonymized Permanent Account Number of the trader, the transaction date, the security ticker, total shares purchased or sold, and the execution price. We limit our analysis to transactions involving stocks included in the Prowess Database (similar to CRSP in the U.S.) maintained by the Centre for Monitoring Indian Economy (CMIE).¹⁵ In addition, we retain only securities that are common shares of domestic stocks and exclude trading activities related to ETFs and foreign stocks.¹⁶ For each retail investor, we further obtain their geographical location at the six-digit PIN Code level, which is essential to match our trading data to information on the location and opening of CAAQMSs.¹⁷ Overall, the initial sample includes equity transactions for 19 million unique investors across India.

We provide the geographic distribution of domestic retail investors across districts, deflated by district population, in Appendix Figure A1. While, as the figure shows, NSE investors are widely distributed across

0–20 kilometers as a distance that would very likely be affected by pollution information from a station (showing also robustness for narrower radii), while in most cases its relevance likely wanes by 40 km.

¹⁴ The National Stock Exchange (India) had the fifth largest market capitalization worldwide and the highest market cap in India as of 31 December 2024. see https://en.wikipedia.org/wiki/List_of_major_stock_exchanges, last accessed April 23, 2025.

¹⁵ Prowess is the standard database employed by researchers studying Indian equity markets. See, e.g., Khanna and Palepu (2000), Goldberg et al. (2010), Balasubramaniam et al. (2023), Bau and Matray (2023).

¹⁶ None of the ETFs in our sample have an explicit ESG orientation.

¹⁷ The PIN Code assigned to an investor is fixed at the time of account registration and remains static in our dataset, which may introduce measurement error. The most likely statistical consequence is attenuation bias (i.e., bias toward zero in coefficient estimates).

the entire country, they are sparser in lower-income provinces, situated in the northeast.¹⁸

We use historical transaction data from as far back as 2004 to construct a holding sample containing an observation for each investor-stock-day following standard methods (e.g., Ben-David and Hirshleifer, 2012; Frydman et al., 2018).¹⁹ We then aggregate investors' portfolio holdings based on their location to match the variation in CAAQMS introduction and conduct our main analysis at the PIN-Code-by-month level. Intuitively, each regional account describes the stock holdings of a representative regional investor.

Identifying Brown Stocks: Our main approach to classifying firms as “brown” or “green” is based on their industry classifications, following (Choi et al., 2020), which in turn uses the definition of the Intergovernmental Panel on Climate Change (IPCC) to classify five sectors as sources of high emissions—Energy; Transport; Buildings; Industry (such as chemicals and metals); and Agriculture, Forestry, and Other Land Use. Then, following (Krey et al., 2014), each sector is classified into sub-categories, each of which is hand matched with industry names provided by Datastream. (Note that air pollution emissions are highly correlated with CO2 emissions through fossil-energy consumption, as noted by Lin et al. (2023) and Agee et al. (2014), making this industry-level measure also a plausible proxy for a firm's contribution to lower air quality.)

We use the industry classifications provided in the Prowess database to hand match to Datastream industries, to classify each firm (via its industry classification) as green or brown based on the Choi et al. (2020) list. Fig. 2 plots the evolution of “brown” and “green” (i.e., non-brown) stocks over our sample period. In Panels A and B we show the number of stocks and market capitalization, and in Panel C we show brown stocks' share of market capitalization. Brown stocks' market share varied between about 30 and 40 percent during 2004–2020, peaking in 2010 then dropping steadily thereafter.

Note that we make this assignment at the industry- rather than firm-level. We do so in part because an industry classification is better-aligned with the likely heuristics used by most retail investors that are our focus, as they are less apt to explore the specific ESG impact of individual firms when making investment decisions.²⁰ This choice is also necessitated by the limited availability of ESG ratings in India, as well as concerns about the reliability of these ratings (Berg et al., 2021, 2022b,a). That is, compared to developed markets like Europe and the United States, the availability and quality of firm-level ESG data in India are relatively sparse and less reliable.²¹

As an alternative approach that focuses more on individual firms, we use Thomson Reuters' “Global 250 Greenhouse Gas Emitters”, which provided the 250 publicly listed companies with the highest greenhouse gas emissions in 2017, which falls during the period we study. There are 12 Indian companies on the list, including Coal India, the world's

highest CO2 emitter.²² Since this list is based on total carbon emissions rather than carbon intensity, it captures size as well as “brownness” which is likely appropriate for the purposes of identifying companies that will be most salient for investors' pollution concerns.

We also observe that while the source of variation in our analysis is based on access to air pollution information, our classifications rely on CO2 emissions. We do so as a matter of expediency—we are able to match all listed firms to brown share classifications using the approach above. While several measures of industry-level air pollutant emissions are available via Trucost, we had limited success in merging our data with Trucost industries—fewer than half of the firms in our sample were matched based on industry name. More importantly, we argue that our CO2-based proxy captures salient information on industries' contribution to air pollution. As noted by Lin et al. (2023) and Agee et al. (2014), both CO2 and air pollution are in large part the result of burning fossil fuels, so the two are highly correlated. We find this to be the case in comparing our own data to Trucost's air pollution measures. For the eighteen industries that can be matched with high confidence (as captured by an industry name similarity score of at least 0.8), the correlation between our indicator variable for a “brown” industry and the Trucost air pollution “impact ratio” is 0.85. (This is even higher than the correlation between our CO2-based measure and Trucost's proxy for greenhouse gas emissions, which is 0.67.)

Pollution Monitoring Stations — CAAQMSs: We obtained geocoded data on the location and timing of continuous ambient air pollution monitoring stations from India's Central Pollution Control Board. The first CAAQMS appeared in 2007, and by 2021 there were 311 stations spread across the country (see Fig. 1, Panel B). We classify a PIN Code as in the “treatment” group if its centroid is located within a 20 kilometer radius of a CAAQMS under the assumption that, when a monitoring station appears, investors in the PIN Code gain ready access to real-time air quality data. We leave a buffer of a “donut hole” region comprised of PIN Codes 20–40 kilometers around the station, and then classify PIN Codes that have centroids located 40–60 kilometers from each station as the “control” group.

We classify a PIN Code as a control only when its centroid is not within the treated region of any other CAAQMS. Thus, the treatment and control groups are mutually exclusive, and the sample is created such that there is no overlap across the groups, with every CAAQMS having a unique set of treated and control PIN Codes. Panel A of Appendix Figure A2 illustrates our assignment approach for the Jodhpur monitoring station, which provides a straightforward case, given that the city hosts only a single station. Panel B shows an example of two monitoring stations in Delhi which illustrates that India's largest cities have multiple stations that are generally within 60 kilometers of each other; this creates some complications in treatment and control assignments, which we address as described above. For this reason, we will also present our results excluding India's 10 largest cities, which are most apt to have such overlaps.

To ensure comparability between treated and control regions, we conduct balance tests across key socioeconomic and investor characteristics prior to the shock, including investor composition (share female, young, and middle-aged, as well as share “sophisticated” and retail share) in Panel A; brown share of investments (overall and by subgroup) in Panel B; quantity of local investors and trading activity in Panel C; economic activity (entry of new firms, nighttime light intensity) in Panel D; pollution and weather in Panel E; and demographics (population, schools per capita, and phone penetration rate) in Panel F. We present these results in Table A1. Columns 1 and 2 report means

¹⁸ To ensure that no particular geography is driving our results, we present a robustness check of our main findings in which we exclude one state at a time. Also note that while only a small fraction of the Indian population actively trade on the NSE, retail investors represent a substantial fraction of trading volume. In our own data, we find that between 37 and 45 percent of trading during our sample period could be attributed to retail traders, a figure that broadly aligns with reporting in the popular press. See, e.g., “Retail investors' rush into India's equity markets has slowed; here is what's happening”, Ashish Rukhaiyar, *Business Today*, May 29, 2023.

¹⁹ Following (Ben-David and Hirshleifer, 2012), we remove a trader's investment from the sample if the cumulative number of shares becomes negative at any point (owing to a purchase that occurred prior to the start of the sample period).

²⁰ Consistent with this assumption, Moss et al. (2023) show that retail investors exhibit muted responses to firms' ESG disclosures, using data from Robinhood.

²¹ For example S&P Global's Trucost database, which we discuss further below, could be matched to fewer than a quarter of the firms in our sample.

²² The full list includes Coal India (rank 1); Reliance Industries (36); NTPC Ltd (54); Oil & Natural Gas Corporation (ONGC, rank 65); Vedanta Ltd (110); Essar Oil (122); Bharat Petroleum Corporation (BPCL) (160); Tata Power Co (184); Adani Power Ltd (208); Hindustan Petroleum Corporation (HPCL) (219); Tata Motors (222); Ashok Leyland (240).

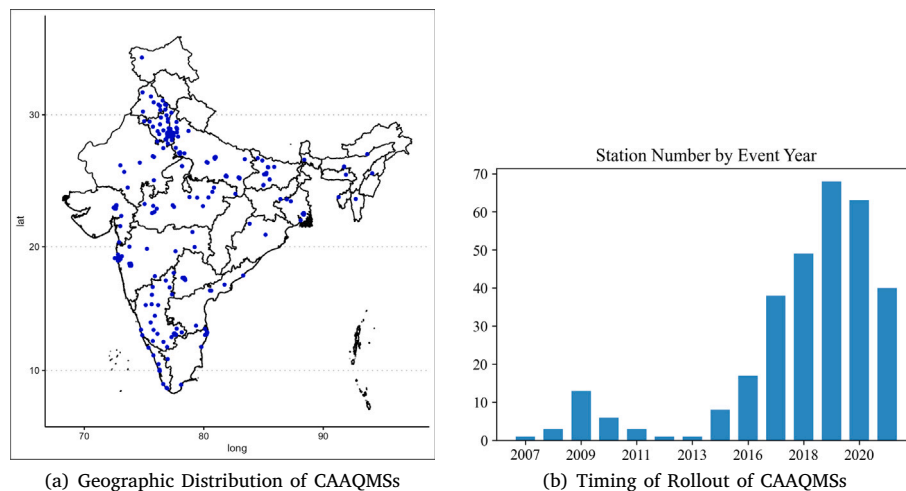


Fig. 1. Geographic distribution and rollout of CAAQMSs.

Note: Panel A plots the geographic locations of the Continuous Ambient Air Quality Monitoring Stations (CAAQMSs) across India. Panel B shows the number of CAAQMSs introduced each year.

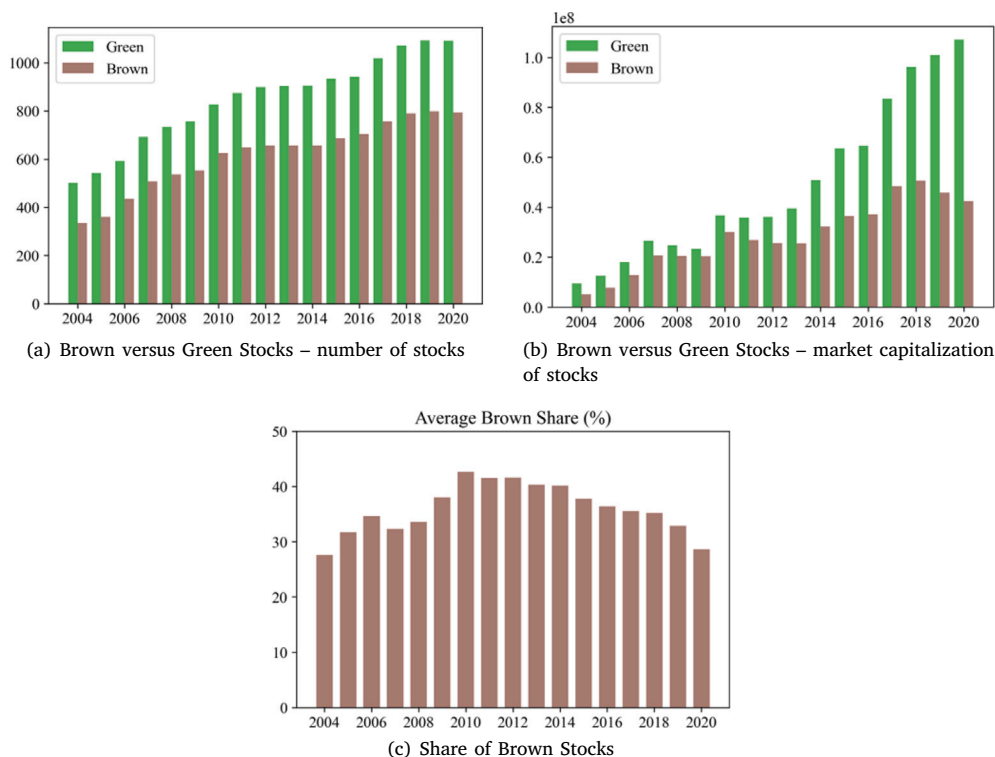


Fig. 2. Distribution of green and brown stocks over Time.

Note: This figure plots the distribution of green and brown stock holding over time. Panel A presents the number of green and brown stocks over time, while Panel B presents the market capitalization of green vs brown stocks by year. Panel C presents the market share of brown stocks (in terms of market capitalization) among all publicly traded equities in India. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

for treated and control PIN Codes, respectively. Since the relevant identifying variation involves a between-PIN Code comparison within a given monitoring station area, in Column 3, we follow [De Janvry et al. \(2023\)](#) to estimate within-station differences using OLS regressions with monitoring station fixed effects. All covariates are balanced across the two regions, suggesting that unobserved confounders are unlikely to systematically bias our triple-differences estimates.

To assess whether endogenous placement of monitoring stations might complicate the interpretation of our results, we also examine the determinants of when a PIN Code is “treated” with a monitoring station. Following [Galiani et al. \(2005\)](#), and using a discrete-time hazard model for a monitoring station’s arrival, we include the same set of covariates in the balance test above, and present these results in Appendix Table A2, for both the full set of treated and control PIN

Table 1
Summary statistics.

	mean	sd	min	p5	p10	p25	p50	p75	p90	p95	max
Brown Share (%)	41.876	9.343	0.000	28.040	32.590	37.670	41.840	45.840	51.190	56.010	100.000
Brown Share Female (%)	41.194	19.085	0.000	4.280	18.080	32.980	41.160	48.400	61.580	76.400	100.000
Brown Share Male (%)	41.935	9.644	0.000	27.640	32.340	37.600	41.900	46.010	51.540	56.580	100.000
Brown Share Young (%)	40.414	17.615	0.000	9.500	20.720	32.030	39.970	47.790	59.270	70.760	100.000
Brown Share MidAged (%)	41.943	11.481	0.000	24.440	30.750	37.180	41.910	46.490	53.100	59.250	100.000
Brown Share Old (%)	42.839	16.066	0.000	15.100	26.390	36.540	42.490	48.450	59.390	70.290	100.000
Pollution AOD	0.543	0.217	0.031	0.264	0.305	0.387	0.505	0.656	0.843	0.954	1.880
No. of Investors (Log)	3.861	1.805	0.693	1.099	1.609	2.485	3.638	4.970	6.541	7.288	10.384
Turnover (Log)	16.846	2.891	0.140	11.407	12.925	15.256	17.195	18.801	20.265	21.028	24.848
Rainfall	3.171	5.322	0.000	0.000	0.000	0.050	0.940	4.170	9.020	13.140	109.590
Temperature	25.764	4.916	3.290	15.750	18.280	23.410	26.620	29.270	31.230	32.500	36.510

Note: This table reports the summary statistics of the key variable used for the empirical analysis in this paper. The summary statistics are for the universe of Indian retail investors who trade on the National Stock Exchange, aggregated at the PIN-Code-month level. We report the average share of brown stocks for retail investors overall, and also disaggregate by gender and age cohort.

Codes (Column 1) as well as just treated PIN Codes (Column 2). We find no significant predictors of the arrival of a monitoring station, mitigating endogeneity concerns.

Pollution Awareness – Web Searches: Our main hypothesis builds on the assumption that easier access to pollution information via CAAQMSs increased the salience of air quality for investors. To examine this assumption, we look at how demand for pollution-related information, as captured by Google keyword searches, changed around the appearance of a monitoring station. To do so, we manually compiled multiple cross-sectional exports from Google Trends to build a complete panel dataset across cities for pollution-related keywords.²³ Our data cover 128 cities from January 2010 to December 2023. To construct our pollution-based index, we combine the popularity indices of four terms: “pollution”, “air quality”, “emissions”, and “industrial pollution”. As a placebo comparison, we created a general political economy search index using the terms “election”, “inflation”, and “growth”. We plot the search index as a function of quarters since the arrival of a monitoring station to align with the timing of our main analyses. As seen in Appendix Figure A3 and Appendix Table A3, we observe a clear upward shift in environmental- and pollution-related searches after a city receives its first monitoring station. No such shift is seen for the placebo search terms. These patterns suggest a greater attention to pollution after the appearance of a monitoring station.

Satellite Data on Pollution: We wish to measure local air pollution at the PIN Code level in a consistent manner both before and after the introduction of CAAQMSs. To do so, we take advantage of data generated by NASA’s Moderate Resolution Imaging Spectroradiometer on its Terra satellite. These readings are used by NASA to generate data on aerosol optical depth (AOD), which is a widely-used proxy for pollution from outdoor particulate matter and reflects the density of various particles such as sulfates, nitrates, black carbon, and sea salts (see, e.g., [Van Donkelaar et al., 2006](#)). Following existing studies such as [Tsai et al. \(2011\)](#), we confirm in Appendix Figure A4 that AOD and PM 2.5 (obtained from the CAAQMSs) are highly correlated, based on a comparison following the implementation of the monitoring program, when both pollution proxies are available. The AOD data are available at a frequency of 30 min in a 10-by-10 kilometer grid. Our main pollution data are average monthly readings for the grid location that contains the PIN Code’s centroid.

One natural concern is the endogenous timing of CAAQMS installations, i.e., monitoring stations may have been placed in areas where pollution is worsening. We discuss this issue in Section 4.2 below, but to briefly summarize, we do not observe any shift in pollution (nor investor attributes or local economic conditions) around the opening of a monitoring station.

Weather Data: Air quality is correlated with sunshine and/or precipitation, which may in turn also affect trading behavior. We thus include meteorology controls for weather. These data are collected from the European Centre for Medium-Range Weather Forecasts (ECMWF) ERA5 Land datasets, which combine observations from satellites, weather stations, and weather forecast models as far back as 1963. As part of the ERA5 climate reanalysis, the data also provide high-resolution information about land surface variables. Similar to the AOD data, we use the zonal statistic method to calculate the monthly average and create a panel data at the PIN-Code-month level for rainfall and for surface temperature, which we include as controls in all of our main specifications.

Sample Description and Summary Statistics: Our regression analysis is at the PIN Code-month level. Our main sample consists of 1849 distinct “treated” PIN Codes and 4845 “control” PIN Codes, for January 2004 to June 2020. The panel is unbalanced—in 65 percent of PIN-Code-month observations, stock holdings are zero so that our main outcome, share of brown stocks, is undefined. Some monitoring rollout occurs during 2017–2019, close to the end of our sample period. To ensure that the underlying sample for each station is a relatively balanced panel (i.e., we do not have observations very distant in time from the appearance of a monitoring station), we have additional filtering, and focus on the ± 4 years time window around the rollout date. The total number of observations for the baseline specification at the PIN Code-month level is 499,036.

Table 1 presents summary statistics for the primary variables we use in our analysis. Aggregated across our entire sample period, we observe that a typical retail investor holds roughly 41.8% of their portfolio in brown stocks, marginally above the overall share of the value of brown stocks in the Indian market (see [Fig. 2](#)). Of relevance for the heterogeneity analyses we present in Section 5.5, Table 1 also provides portfolio brown shares across investor groups. We observe that female investors hold a marginally lower fraction of assets in brown stocks relative to their male counterparts (41.2% versus 41.9%), while the share of brown stocks is notably lower for young investors (40.4%), relative to middle-aged and old investors (41.9% and 42.8% respectively).

4. Empirical strategy

4.1. Empirical specifications

Our objective is to assess how the sensitivity of an individual’s green investments to pollution changes after the monitoring program improves public access to air quality information. Specifically, we investigate the connection between pollution exposure and investments in brown stocks, which we refer to as the brown-share-pollution gradient, and measure how this gradient differs before versus after the (staggered) rollout of local monitoring stations across regions. As described in Section 2, our methodology differs slightly from that

²³ Google Trends data are available only at the city level, a much higher level of aggregation than a PIN Code.

of Barwick et al. (2019), since we do not rely exclusively on the staggered rollout of monitoring stations for identification. In addition, to control for any time-varying local confounding factors, we leverage detailed information on investors' geographies and add a third layer of differences. Specifically, we compare changes in outcomes for retail investors close to a monitoring station (within 20 kilometers, the *treated ring*) with those located further away (40–60 kilometers, the *control ring*).

We estimate the change in the brown-share-pollution gradient using the following specification:

$$\begin{aligned} \text{Brown Share}_{p(m),t} = & \beta \times \text{Treat}_p \times \text{Pollution}_{p,t} \times \text{Post}_{m,t} \\ & + \sum_k \alpha_k \times \text{Other Interactions} + \mathbf{X}'_{p,t} \boldsymbol{\theta} \\ & + \gamma_p + \lambda_{m,t} + \varepsilon_{p,t} \end{aligned} \quad (1)$$

where p, m, t denote PIN Code, monitoring station, and month, respectively. $\text{Brown Share}_{p(m),t}$ represents the average value of brown stocks summed across all retail investors in each PIN Code divided by the total value of retail investors' portfolios in PIN Code p in monitoring station area m at time t . A station area m contains PIN Codes with centroids within 20 kilometers of the monitoring station — treatment PIN Codes — and PIN Codes 40–60 kilometers from the station — control PIN Codes. Treat_p is an indicator variable that equals one if a PIN Code falls into the treated group (i.e., within 20 kilometers of the monitoring station) and zero otherwise. The variable, $\text{Post}_{m,t}$, captures exposure to this information in real-time as a result of the arrival of a monitoring station. $\text{Pollution}_{p,t}$ is the local ambient air pollution measure from AOD in PIN Code p and month t . It is worth reiterating that AOD measures local residents' exposure to pollution both pre- and post-arrival of a monitoring station, although air quality information (collected by the station) is available only in the post period. *Other Interactions* include all lower-order terms, such as $\text{Treat}_p \times \text{Pollution}_{p,t}$, $\text{Treat}_p \times \text{Post}_{m,t}$, $\text{Post}_{m,t} \times \text{Pollution}_{p,t}$, and Treat_p , $\text{Pollution}_{p,t}$ and $\text{Post}_{m,t}$. $\mathbf{X}_{p,t}$ is a vector of control variables including local trading activities (number of retail investors, total turnover) and weather conditions (rainfall, temperature) in PIN Code p and month t .

The set of PIN Code fixed effects, γ_p , controls for time-invariant factors of a PIN Code, enabling us to identify differences based on time series variation in whether a nearby monitoring station has opened. To account for time-varying characteristics in the region around monitoring stations, we include station area $\times$ time fixed effects ($\lambda_{m,t}$) in our most saturated specification. This allows us to identify differences between treatment and control groups for each station at the same point in time. We cluster standard errors at the PIN Code level to account for possible serial correlations among the sample periods. The coefficient of interest, β , captures the changes in the relationship between pollution exposure and investment in brown stocks that occur with the appearance of a monitoring station, comparing nearby (treated) versus more distant (control) PIN Codes.

We also implement an event-study version of the specification in Eq. (1) to explore dynamics around the treatment date. To do so we substitute for the $\text{Post}_{m,t}$ dummy with a collection of indicators measuring the quarters elapsed since the installation of a monitoring station:

$$\begin{aligned} \text{Brown Share}_{p(m),t} = & \sum_{i=-6}^6 \beta_i \times \text{Treat}_p \times \text{Pollution}_{p,t} \times D(t \in i)_{m,t} \\ & + \sum_k \alpha_k \times \text{Other Interactions} + \mathbf{X}'_{p,t} \boldsymbol{\theta} \\ & + \gamma_p + \lambda_{m,t} + \varepsilon_{p,t} \end{aligned} \quad (2)$$

where $D(t \in i)_{m,t}$ is a dummy variable that takes a value 1 if the month t belongs to the k th quarter since the monitoring station appeared. This specification allows for the tracking of the evolution of β_i over time in relation to the rollout of monitoring stations and provides

a graphical representation of the dynamics of brown-share-pollution gradients. The number of pre- and post-treatment periods available varies across different waves of PIN Codes. To ensure a balanced time frame for our analysis, we report β_i for a period of 6 quarters prior to and following the arrival of a monitoring station, while binning the remaining more distant periods in the sample.

4.2. Discussion of possible threats to identification

Most previous studies have focused on estimating the pollution-concentration-response function, which quantifies the causal effect of air pollution exposure on health and non-health outcomes. However, identifying this relationship is challenging due to the endogeneity of ambient pollution levels. Endogeneity can arise from omitted variables, such as individual endogenous location choice, or measurement errors. Graff Zivin and Neidell (2013) provide a detailed review of the sources of endogeneity. Eq. (1) emphasizes that our focus is distinct from prior work. We do not aim to estimate the impact of pollution *per se*, but rather to measure how the causal effect (sensitivity of brown share to pollution exposure) varies before versus after the introduction of monitoring stations.

When attempting to estimate the slope of the relationship between brown shares and pollution separately using data from before and after the station rollout, the endogenous nature of pollution could lead to inconsistent estimates for both periods. However, assuming certain conditions hold, the inconsistency in slope estimates could offset each other, hence making the OLS estimate of β in Eq. (1) consistent. The crucial identification assumption is that the rollout schedule is not correlated with the difference in time-varying unobservables between treated and control areas that could drive the observed outcome, given the controls.

In this section, we investigate the validity of the identifying assumption and consider the potential estimation biases that arise from a staggered rollout setting, with reference to the recent literature on difference-in-differences methodologies.

CAAQMS Rollout: Although the assignment of CAAQMS is not entirely random, it is primarily the result of negotiations between the Central Pollution Control Board and state-level pollution control authorities. As discussed in Section 2, the assignment of CAAQMS is influenced by a long list of exogenous concerns, including population, geographic obstruction, power availability, and so forth. Note that we focus on a relatively narrow geographical region and examine geographically proximate control and treatment PIN Codes around the same monitoring station. Moreover, our regression analyses include station-area $\times$ time fixed effects, and therefore eliminate any potential threats to our identification posed by time-varying differences in local characteristics. Thus, we argue that the location and timing of monitoring are largely exogenous to the portfolio allocation decisions of retail investors.

To probe empirically whether endogenous rollout is a substantial concern, we first test whether differences in time-varying local characteristics between control and treatment PIN Codes correlate with the policy rollout, after controlling for the fixed effects included in Eq. (1). We begin by plotting the dynamics of pollution levels (as captured by AOD) between treated and control areas in Fig. 3, as a function of time relative to monitoring. We do not observe any significant shift in the local pollution gap between treated and control investors before versus after the introduction of a monitoring station. Second, we examine a set of confounding factors, including local investor composition as captured by growth of investor base and entry of new investors, and economic activities (entry of new firms and nighttime light intensity) that may differentially impact the portfolio allocation of individual investors. We show in these “balance” tests that program rollout is uncorrelated with changes in these attributes (See Appendix Table A5), supporting the identification assumption described at the beginning of Section 4.2. Finally, the event-study analysis detailed in Eq. (2)

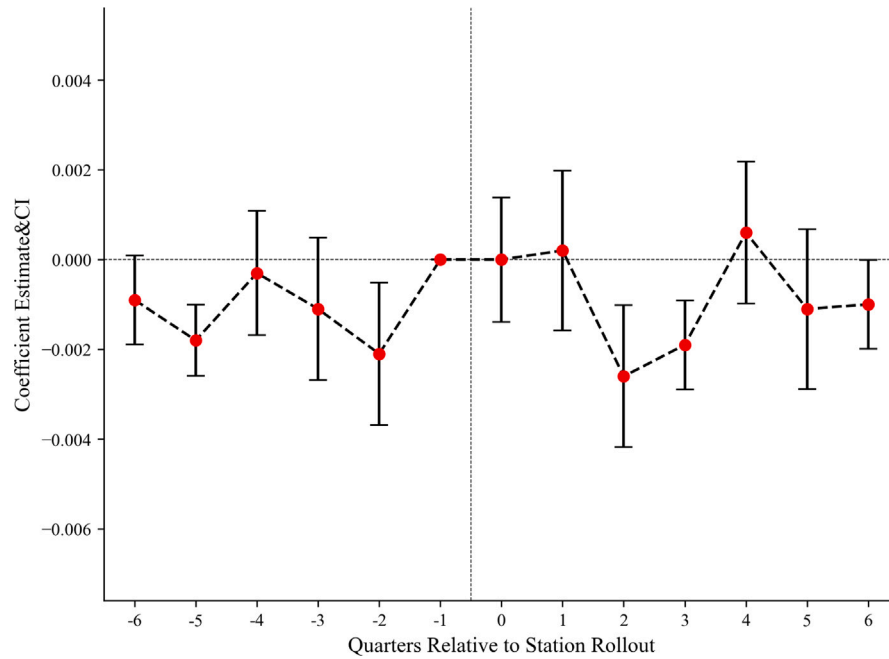


Fig. 3. Pollution levels around CAAQMS installation.

Note: This figure plots the trend in pollution following the installation of Continuous Ambient Air Quality Monitoring Stations. Specifically we plot the coefficients $\{\beta_i\}$ from the following specification:

$$Pollution_{p(m),t} = \sum_{i=-6}^6 \beta_i \times Treat_p \times D(t \in i)_{m,t} + \mathbf{X}'_{p,t} \boldsymbol{\theta} + \gamma_p + \lambda_{m,t} + \varepsilon_{p,t}$$

where $Pollution_{p(m),t}$ denotes the average pollution in PIN Code p belonging to a monitoring station area m at time t . A station area m includes treated PIN Codes (those within 20 kilometers of the station) and also control PIN Codes (40–60 kilometers from the station). $Treat_p$ is an indicator variable which denotes PIN Codes that are in the treated group. $\mathbf{X}_{p,t}$ is a set of controls for PIN Code p at time t , including local trading activities (number of retail investors, total turnover by retail investors) and weather conditions (rainfall, temperature). γ_p is a set of PIN Code fixed effects and $\lambda_{m,t}$ is a set of station area $\times$ time fixed effects. The graph shows the estimated coefficients $\{\beta_i\}$ as well as 95% confidence intervals obtained from standard errors clustered at the PIN Code level.

helps identify possible endogeneity issues, to the extent that they are reflected in pre-existing trends in the outcome variable. We return to this issue below in the discussion of Fig. 5, which will lend additional support to the identification assumption.

Bias in Staggered DID: A recent literature on difference-in-differences estimation indicates that the two-way-fixed-effects (TWFE) estimator may be thought of as a weighted average of all potential 2×2 DID estimates among three groups, and can be biased when there are heterogeneous treatment effects over time or across units (Goodman-Bacon, 2021; Callaway and Sant’Anna, 2021; Sun and Abraham, 2021). This bias results from some of the 2×2 DID estimates entering the average with negative weights, thus introducing biased estimates that dilute the true treatment effect. The “negative weight” problem is primarily driven by comparisons between previously treated groups as controls and newly treated groups.

Our results are not affected by the estimation concern mentioned above, as we are interested in the brown-share-pollution gradient rather than the level of brown share itself. Our specification can be viewed as a “stacked” difference-in-differences model, which produces an efficient estimator for uncovering the aggregated treatment effect through OLS (Cengiz et al., 2019; Baker et al., 2022). In essence, we estimate the average treatment effect on the pollution gradient across multiple “canonical” DID regressions with two groups and clean pre- and post-periods.

To explore the analogy of TWFE in our setting, below we also estimate the event study specification separately for investors located in the treated and control PIN Codes and plot the β_i estimates. As we discuss further in presenting our main results, we observe a significant shift in the brown-share-pollution gradient, but only for investors in treated PIN Codes, lending further support to the validity of the parallel trends assumption.

5. Results

5.1. Visual evidence

We begin with a visual presentation of how the brown-share-pollution gradient shifts with the introduction of local monitoring stations. We present our results in a canonical 2×2 difference-in-differences setup, which is relatively transparent and easy to interpret, and later also show event-study plots to highlight that these shifts coincide precisely with the arrival of monitoring stations.

In each panel of Fig. 4, we show a binned scatterplot that displays the relationship between observed air quality and brown share holdings for both control and treated PIN Codes (top and bottom) before and after the arrival of a monitoring station (left and right). We sort PIN-Code-month average pollution into 100 bins and plot average pollution against brown share. We adjust the local brown share for the time-varying mean in a station area across both control and treated PIN Codes. Consistent with a lack of attention to air quality, we observe no correlation between brown share and air quality before the appearance of a monitoring station in both treated and control PIN Codes (left two panels). We similarly observe no relationship in control PIN Codes after a monitoring station appears (upper right panel). We observe a distinct pattern in the lower right panel which shows the relationship for nearby (treated) PIN Codes in the post period—in this case there is a clear negative relationship between air quality and the investors’ brown share holdings.

5.2. Baseline estimation

Both to control for a wider array of geographic and temporal attributes, as well as to study how the pollution-brown-share gradient

Table 2

The impact of air quality information via CAAQMS on the brown-share-pollution gradient.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dep. Var. = Brown Share (%)								
Pollution × Treated × Post	−1.5319** (0.7715)	−2.1017** (0.8397)	−1.7251* (0.9762)	−2.5007*** (0.9617)	−1.3634* (0.7675)	−2.0087** (0.8365)	−1.5687 (0.9739)	−2.4050** (0.9607)
Pollution × Treated	0.7576* (0.3914)	1.1183** (0.4283)	0.9427* (0.4958)	1.2966*** (0.4835)	0.6620* (0.3893)	1.0331** (0.4270)	0.8627* (0.4945)	1.2234** (0.4826)
Pollution × Post	0.2348 (0.5419)	0.8557 (0.6750)	0.7933 (0.6974)	2.0042 (1.6520)	−0.0026 (0.5387)	0.7614 (0.6729)	0.7592 (0.6951)	1.8593 (1.6451)
Treat × Post	0.9842*** (0.2711)	0.6365** (0.2733)	0.5494 (0.3393)	0.6440** (0.3027)	0.5404** (0.2670)	0.3639 (0.2704)	0.3871 (0.3367)	0.3974 (0.2995)
Pollution	−0.2475 (0.2557)	−0.6442* (0.3894)	−0.4243 (0.7409)	−1.0167 (0.7642)	−0.1799 (0.2530)	−0.5311 (0.3883)	−0.4084 (0.7372)	−0.9371 (0.7607)
Post	−0.2702 (0.1658)	−0.0112 (0.1946)	−0.0175 (0.2513)		−0.1455 (0.1645)	0.0196 (0.1942)	−0.0172 (0.2509)	
No. of Investors (Log)					−2.1658*** (0.1905)	−1.6583*** (0.1987)	−1.3128*** (0.2034)	−1.5185*** (0.2034)
Turnover (Log)					−0.0971** (0.0406)	−0.1069*** (0.0407)	−0.0955** (0.0413)	−0.1023** (0.0409)
Rainfall					−0.0024 (0.0040)	−0.0079 (0.0054)	−0.0011 (0.0084)	−0.0114 (0.0101)
Temperature					0.0100 (0.0066)	−0.0561* (0.0312)	−0.1447** (0.0572)	−0.0566 (0.0734)
Pincode	Y	Y	Y	Y	Y	Y	Y	Y
Year-month	Y				Y			
State × Year-month		Y				Y		
District × Year-Month			Y				Y	
Station × Year-month				Y				Y
Observations	499,036	499,036	488,681	498,310	499,036	499,036	488,681	498,310
R-squared	0.537	0.546	0.599	0.560	0.540	0.548	0.600	0.561

Note: This table shows the elasticity of the share of brown stocks held in investors' portfolios with respect to information on pollution following the installation of Continuous Ambient Air Quality Monitoring Stations. Specifically we report the coefficient β from the following specification:

$$\text{Brown Share}_{p,m,t} = \beta \times \text{Treat}_p \times \text{Pollution}_{p,t} \times \text{Post}_{m,t} + \sum_k \alpha_k \times \text{Other Interactions} + X'_{p,t} \theta + \gamma_p + \lambda_{m,t} + \varepsilon_{p,t}$$

where $\text{Brown Share}_{p,m,t}$ denotes the average share of brown stocks of retail investors in PIN Code p belonging to station area m at time t . A station area m includes treated PIN Codes (those within 20 kilometers of the station) and also control PIN Codes (40–60 kilometers from the station). Treat_p is an indicator variable which denotes PIN Codes that are in the treated group. The binary variable $\text{Post}_{m,t}$ is equal to one after the installation of a local monitoring station. We demean the pollution variable for ease of interpretation of lower-order coefficients. $X_{p,t}$ is a set of controls for PIN Code p at time t , including local trader and weather characteristics (see text for details). γ_p is a set of PIN Code fixed effects and $\lambda_{m,t}$ is a set of station area $\times$ time fixed effects. Columns 1–4 report results without controls $X_{p,t}$ while columns 5–8 report results with these controls. Standard errors are clustered at the PIN Code level.

*** Indicate statistical significance at the 1% level.

** Indicate statistical significance at the 5% level.

* Indicate statistical significance at the 10% level.

evolves over time, we present an event-study plot of β_i in Fig. 5. The value of the coefficient for the quarter just before the event is set as the benchmark and normalized to zero. The vertical axis provides the estimated change in the brown-share-pollution gradient ($\frac{\partial \text{Brown Share}}{\partial \text{Pollution}}$), and the horizontal axis provides the quarters relative to the monitoring program. We observe two patterns. First, for the quarters leading up to the introduction of CAAQMSs, the β_i estimates are statistically indistinguishable from zero, lending support to the validity of the parallel trend assumption. Second, the brown-share-pollution gradient becomes negative immediately after the monitoring program rollout, with the estimated β values between -2 and -3 in the post-periods.

Table 2 presents our baseline regression results based on Eq. (1), in which we test the robustness of the patterns described above using different sets of fixed effects and controls (note that for ease of interpretation, we demean the pollution variable in all specifications). We present our most parsimonious model in Column 1, with PIN Code and year-month fixed effects. Columns 2 and 3 include interactive state-time and district-time fixed effects respectively, to absorb time-varying confounding effects at the regional level (e.g., changes in local economic conditions or employment) and to allow for common trends specific to each region. Column 4 is the most stringent specification, including interactive monitoring-station-time fixed effects and

PIN Code fixed effects.²⁴ Columns 5–8 provide analogous specifications, but include additional controls for local trading activity and weather. As explained in the preceding section, our primary interest is in the coefficient on the three-way interaction of $\text{Treat}_p \times \text{Pollution}_{p,t} \times \text{Post}_{m,t}$. This reflects the differential response of investors to pollution after a monitoring station appears, in treated relative to control PIN Codes. The point estimate of -2.4 in our favored specification in Column 8 implies that, relative to the pre-period, after a monitoring station appears, an increase in pollution from the 25th percentile to the 75th percentile (an AOD increase of 0.27, which is the within-station interquartile range) is associated with a decline of the brown share in investors' portfolios in treated areas by 0.65 percentage points (0.27×2.4). This represents a 1.6 percent decrease relative to the mean brown share of 41 percent. In other words, the brown-share-pollution elasticity becomes 3.2 percentage points more negative once a monitoring station appears.²⁵

²⁴ These high-dimensional fixed effects essentially convert the model to a stack of “canonical” DID regressions with unit fixed effects and time fixed effects.

²⁵ The estimated change in brown-share-pollution elasticity is $\frac{0.65/41.87}{0.27/0.54} = 3.2\%$.

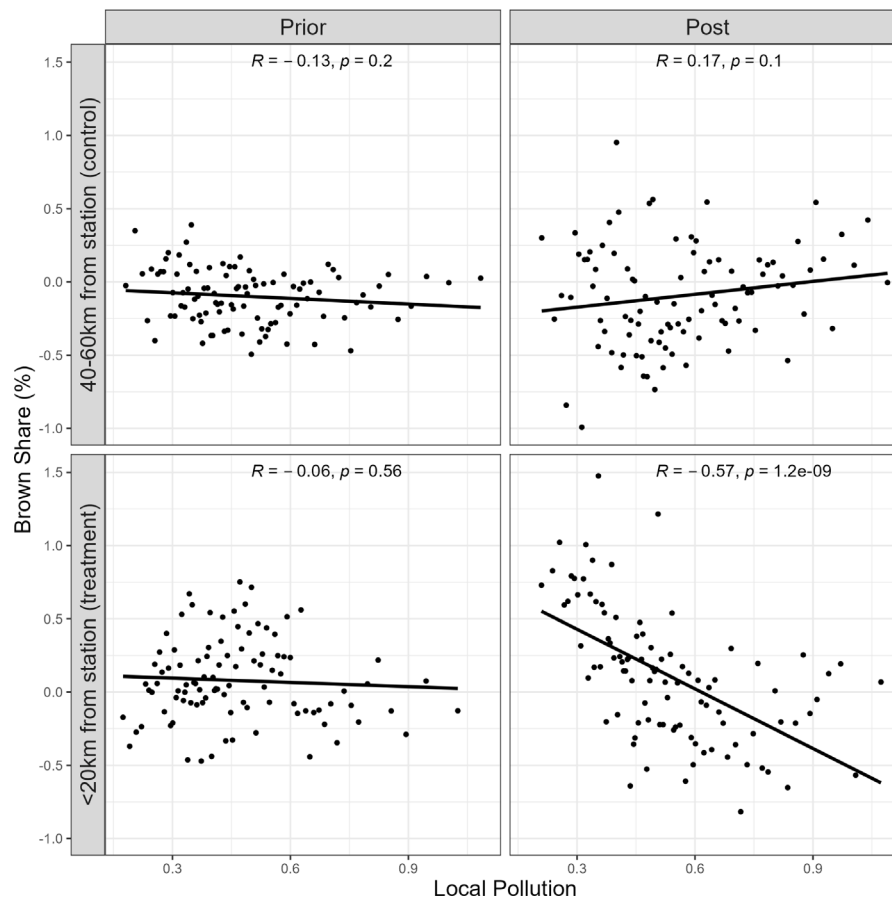


Fig. 4. Correlation of brown share and pollution – control versus treated PIN codes.

Note: This figure presents scatter plots relating the share of brown stocks and local pollution for both the control and treated groups before and after the installation of Continuous Ambient Air Quality Monitoring Stations. We sort pollution rates into 100 bins and plot average pollution against the average brown share holdings. The brown share is adjusted for the time-varying mean in a station area for both the control and treatment groups. The top two panels represent the control PIN Codes, while the bottom two panels represent the treated PIN Codes. The left two panels show the relationship prior to the installation of CAAQMS, while the right two panels show the relationship post-installation.

Table 2 also demonstrates that retail investors' investments in brown stocks in control PIN Codes are unaffected by pollution both before and after the introduction of a monitoring station—that is, the coefficients on the *Pollution* and *Pollution × Post* terms are both insignificant when *Treated* = 0. This finding helps to validate our treated and control assignments, and mitigates potential concerns over spatial spillovers.

To further highlight the fact that our findings are driven by shifts in treated PIN Codes around the appearance of monitoring stations, we show event-study plots for treated and control investors separately. Specifically, in Fig. 6 we show the point estimates and 95% confidence intervals for the coefficients generated by a variant of specification (2) above, in which we split the sample into investors located in treated and control PIN Codes.

From an identification perspective, it is comforting that our measure of brown share investments is essentially flat in the two years (eight quarters) preceding the installation of a monitoring station for both treated and control PIN Codes. At that point there is a clearly discernible shift in the sensitivity of brown investment share to AOD in treated PIN Codes—higher AOD (i.e., worse pollution) is associated with a lower brown share after a monitoring station arrives. We observe no such shift in control PIN Codes. These patterns also mitigate a concern suggested by the lower-order terms in Table 2—that treated PIN Codes may have had a lower sensitivity to pollution in treated PIN Codes before the appearance of a monitoring station. This may result from the structure imposed by the triple-interaction specification,

which generally forces fixed effects and controls to have identical coefficients for treatment and control investors. Our event plot suggests that this difference disappears when we allow for the more flexible specification illustrated in the event plot (and similarly in the alternative approach we introduce at the beginning of the next section, which accounts for a bias resulting from staggered rollout, we do not observe a pre-CAAQMS difference).

Selling Brown vs Buying Green: If investors in our sample are simply trading one set of shares for another, we would necessarily see a symmetry between the sale of brown stocks and the purchase of green ones. To the extent investors instead divest themselves of brown stocks without reinvesting, or alternatively purchase green stocks with outside funds, our main findings could be the result of either selling or buying behavior. Our prior is that the former would more plausibly drive the results, as the “brown” industries in our sample—coal; oil and gas; utilities; transportation; and so forth—are linked more directly to pollution than the green industries that are simply their complement (insurance; banking; services in general).

We assess retail investors' directional trades on brown (and green) stocks using order imbalance measures (Kaniel et al., 2008; Boehmer et al., 2021). Specifically, for days with high and low pollution, we compute the net trading (i.e., net buy) of brown and green stocks separately, by subtracting the value of shares sold by individuals from the value of shares bought, and standardize the measure by the total daily dollar volume. We define order-imbalance of brown stocks as follows, using trading data from high-pollution days (those with pollution levels

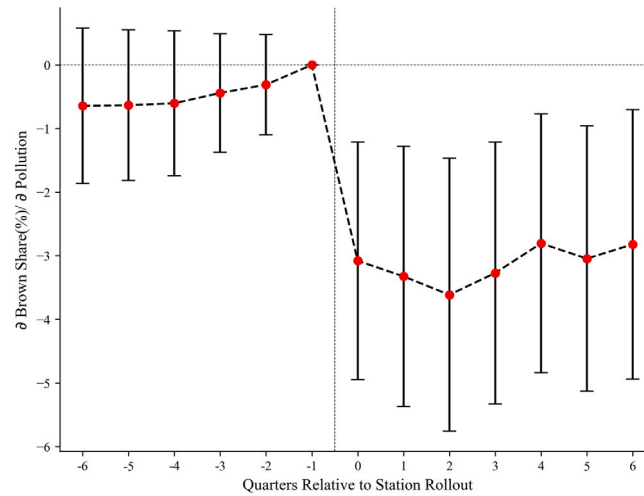


Fig. 5. Sensitivity of brown share to pollution.

Note: This figure plots the sensitivity of brown investment to local pollution before and after the installation of Continuous Ambient Air Quality Monitoring Stations. Specifically we plot the coefficients $\{\beta_i\}$ from the following specification:

$$\text{Brown Share}_{p(m),t} = \sum_{i=-6}^6 \beta_i \times \text{Treat}_p \times \text{Pollution}_{p,t} \times D(t \in i)_{m,t} + \sum_k \alpha_k \times \text{Other Interactions} + \mathbf{X}'_{p,t} \boldsymbol{\theta} + \gamma_p + \lambda_{m,t} + \varepsilon_{p,t}$$

where $\text{Brown Share}_{p(m),t}$ denotes the average share of brown stocks held by retail investors in PIN Code p belonging to a station area m at time t . A station area m includes both treated (those within 20 kilometers of the station) and control PIN Codes (40–60 kilometers from the station). Treat_p is an indicator variable which denotes PIN Codes that are in the treated group. $\mathbf{X}_{p,t}$ is a set of controls for PIN Code p at time t , including local trader and weather characteristics (see text for details). γ_p is a set of PIN Code fixed effects and $\lambda_{m,t}$ is a set of station area $\times$ time fixed effects. The graph shows the estimated coefficients $\{\beta_i\}$ as well as 95% confidence intervals obtained from standard errors clustered at the PIN Code level.

above the median of distribution for all PIN-Code-days):

$$\text{OI}_{\text{Brown}}^{\text{High Pollution}} = \frac{\text{Value of Brown Stocks Purchased}}{\text{Total Value Traded}} - \frac{\text{Value of Brown Stocks Sold}}{\text{Total Value Traded}} \quad (3)$$

We may produce an exactly analogous measure for low-pollution order imbalance, using the same calculation for trading on low-pollution days (and then repeat the exercise to calculate high- and low-pollution order imbalance for green stocks). We then focus on the ratio of $\text{OI}_{\text{High Pollution}}^{\text{High Pollution}}$ to $\text{OI}_{\text{Low Pollution}}^{\text{Low Pollution}}$, which captures the difference in directional (i.e., net purchase) trading by local investors between high- and low-pollution days and examine its changes before and after the arrival of a monitoring station. Specifically, we estimate a difference-in-difference specification as follows for brown and green stocks separately.

$$\frac{\text{OI}_{\text{High Pollution}}^{\text{High Pollution}}}{\text{OI}_{\text{Low Pollution}}^{\text{Low Pollution}}} = \beta \times \text{Treat}_p \times \text{Post}_{m,t} + \mathbf{X}'_{p,t} \boldsymbol{\theta} + \gamma_p + \lambda_{m,t} + \varepsilon_{p,t} \quad (4)$$

Fig. 7 presents the estimation results based on Eq. (4). Each line of each panel, from left to right, displays results starting from the most parsimonious model up to the most stringent specification with interactive monitoring-station-time fixed effects and PIN Code fixed effects. As may be readily seen in the left half of the figure, retail investors divest from high-emission industries on highly polluted days (compared to those in control regions) following the appearance of a monitoring station. By contrast, the right panel reveals no such effect for green stocks.

5.3. Robustness and interpretation

Alternative Definition of Brown Stocks: As an alternative proxy for whether a firm is brown, we focus on the most extreme polluters listed on the NSE based on the Thomson Reuters' list of the "Global 250 Greenhouse Gas Emitters", as described in Section 3. These twelve companies are the largest and most visible polluters in India that are plausibly the most salient link to air pollution in investors' minds. In

Appendix Table A6, we re-estimate our main analyses from Table 2, defining brown share instead as the fraction of value held in these firms. Our findings are directionally the same as those we report in our main results, but the implied effect is of much larger economic magnitude. For example, in our preferred specification in Column 8, the coefficient of primary interest is -0.833 . This estimate implies that an increase in pollution from the 25th percentile to the 75th percentile (an AOD increase of 0.27, the within-station interquartile range) is associated with a decline in the brown share in investors' portfolios in treated areas of 0.225 percentage points (0.27×0.833), representing a 4.88% decrease relative to the mean portfolio share of these companies of 4.61%; this is an effect size that is roughly three times what we document for the broader brown share definition.

Alternate Definitions of Control and Treatment Groups: As discussed in Section 4.2, recent advances in applied econometrics have shown that, when employed in staggered difference-in-differences settings, two-way fixed effects models may yield biased coefficient estimates due to varying treatment effects across time or units. As noted earlier, we believe that such concerns are less applicable to our setting, as our primary focus is on the brown-share pollution gradient rather than the brown share level. Further, our baseline specification uses a stacked difference-in-difference by using a control group of more distant PIN Codes for every treated (nearby) PIN Code of a monitoring station. Nonetheless, to further assess the robustness of our results, we adopt the stacked method outlined in prior research (Cengiz et al., 2019; Baker et al., 2022) and make comparisons of treated PIN Codes versus not-yet-treated and/or never-treated PIN Codes, defined based on the timing of their treatment.

The basic idea is to create, for each treated cohort, "clean" datasets that combine the "treated" PIN Codes (those within 20 kilometers of the station) with other nearby PIN Codes that serve as controls. We consider two different approaches to choosing the "clean" controls within the treatment window for each treated cohort: (1) We include only observations for never-treated PIN Codes, i.e., those that were not treated during our sample period, as comparison groups in a given

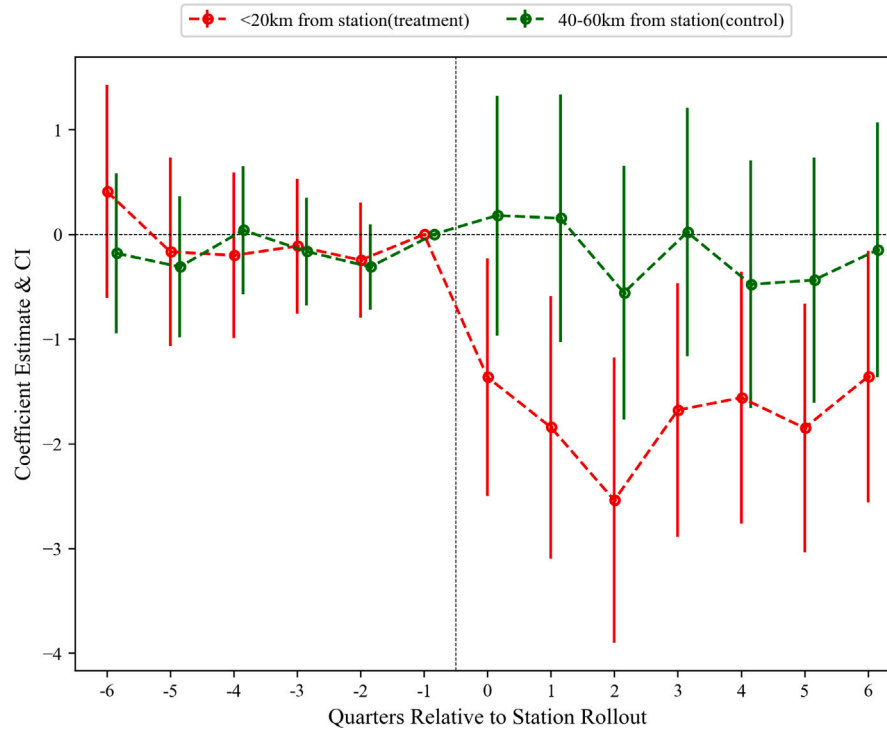


Fig. 6. Sensitivity of brown share to pollution – control versus treated.

Note: This figure plots the sensitivity of brown investment to pollution before and after the installation of Continuous Ambient Air Quality Monitoring Stations separately for the control and treatment groups. Specifically, we plot the coefficients $\{\beta_i\}$ from the following specification separately for the control and treatment groups:

$$\text{Brown Share}_{p(m),t} = \sum_{i=-6}^6 \beta_i \times \text{Pollution}_{p,t} \times D(t \in i)_{m,t} + \sum_k \alpha_k \times \text{Other Interactions} + \mathbf{X}'_{p,t} \boldsymbol{\theta} + \gamma_p + \lambda_t + \varepsilon_{p,t}$$

where $\text{Brown Share}_{p(m),t}$ denotes the average share of brown stocks of retail investors in PIN Code p belonging to a station area m at time t . A station area m includes treated PIN Codes (those within 20 kilometers of the station) and also control PIN Codes (40–60 kilometers from the station). $\mathbf{X}_{p,t}$ is a set of controls for PIN Code p at time t , including local trader and weather (see text for details). γ_p is a set of PIN Code fixed effects and λ_t is time fixed effects. The graph shows the estimated coefficients $\{\beta_i\}$ as well as 95% confidence intervals obtained from standard errors clustered at the PIN Code level.

event window; (2) we include “not-yet-treated” PIN Codes, defined as those that do not get treated within 4 years for each treated cohort, in addition to the never-treated units.

We then stack the resultant datasets and estimate the following specification, which is similar to the baseline analysis:

$$\text{Brown Share}_{p,c,t} = \beta \times \text{Treat}_p \times \text{Pollution}_{p,t} \times \text{Post}_{c,t} + \sum_k \alpha_k \times \text{Other Interactions} + \mathbf{X}'_{p,t} \boldsymbol{\theta} + \gamma_{p,c} + \lambda_{c,t} + \varepsilon_{p,c,t} \quad (5)$$

where p , c , t denote a PIN Code, cohort (dataset) and year-month, respectively. $\gamma_{p,c}$ and $\lambda_{c,t}$ represent the PIN Code $\times$ Cohort and Cohort $\times$ Year-Month fixed effects. When controlling for Cohort $\times$ Year-Month fixed effects, we ensure that the coefficients are estimated by comparing treated PIN Codes solely to “clean” controls in their respective dataset. Appendix Table A7 presents the results, which show that the brown-share-pollution gradient across specifications ranges from -3.81 to -1.86 , estimates that are broadly in line with our baseline findings.

Finally, turning to alternative approaches to treatment group definition, in Appendix Table A8, we present results based on a “treated” group definition as investors within 15 or 10 or 5 kilometers of a monitoring station; the point estimates in our preferred within-station specification are virtually unchanged.

Sensitivity to Alternative Sample Exclusions: Appendix Table A9 excludes PIN Codes in the largest metropolitan areas since, as explained in Section 2, these PIN Codes offer a less straightforward delineation of

treatment and control assignments. Specifically, we exclude the 10 most populous urban areas in India (Delhi, Mumbai, Kolkata, Bangalore, Chennai, Hyderabad, Ahmedabad, Surat, Pune, and Jaipur). We observe substantially larger point estimates on the three-way interaction relative to our main results. We also conduct a robustness check to assess whether our results are driven by any single state. To do so, we re-estimate the main effect from our preferred specification of Column 8 in Table 2 while excluding one state at a time. These results are presented in Appendix Figure A5, and show that our findings are robust to the removal of any particular state.

Additional Stock Characteristics: A potential concern in interpreting our results is a possible correlation between a stock’s “greenness” and other company-level attributes. While it is not obvious *ex ante* why investor response to pollution should involve other attributes (and as such these variables could be seen as bad controls), we nonetheless consider the possibility that the changes in green investment that we document may reflect shifts in portfolios along other dimensions. To address this concern, we construct portfolio shares based on other stock traits, including size, age, and market-to-book, and estimate the impact of pollution monitoring stations on the elasticity of portfolio shares with respect to pollution exposure. We adopt a similar approach to the one used in constructing our primary variable, “brown share”. For example, we first define value/growth stocks as those with a high/low book-to-market ratio (above/below the median of the cross-sectional distribution at year $t - 1$) and then compute the average share of value stocks held by retail investors in PIN Code p at time t . We report the

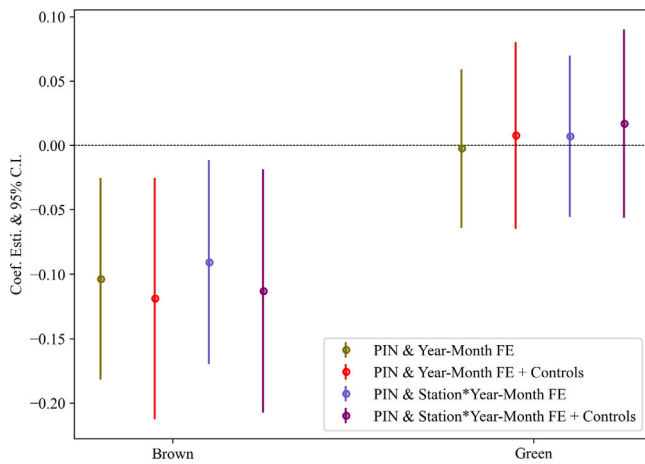


Fig. 7. Selling brown versus buying green.

This figure presents the results on whether the increased sensitivity of local brown shares to air pollution after the opening of nearby monitoring station is driven by (1) brown divestment, whereby investors sell companies with high emissions; or (2) green investment, whereby investors buy firms with low emissions; or (3) both brown divestment and green investment. Specifically, we plot the coefficient estimates along with 95% confidence intervals on $Treat_p \times Post_{m,t}$ from the following specification (or its variants), separately for brown and green stocks:

$$\frac{OI_{High\ Pollution} - OI_{Low\ Pollution}}{p(m),t} = \beta \times Treat_p \times Post_{m,t} + X'_{p,t} \theta + \gamma_p + \lambda_{m,t} + \varepsilon_{p,t}$$

where $\frac{OI_{High\ Pollution} - OI_{Low\ Pollution}}{p(m),t}$ denotes the difference in order imbalance (i.e., net purchases) for trading by local investors between high- and low-pollution days. A station area m includes treated PIN Codes (those within 20 kilometers of the station) and also control PIN Codes (40–60 kilometers from the station). $Treat_p$ is an indicator variable which denotes PIN Codes that are in the treated group. $X_{p,t}$ is a set of controls for PIN Code p at time t , including local trader and weather characteristics. γ_p is a set of PIN Code fixed effects and $\lambda_{m,t}$ is a set of station area $\times$ time fixed effects. The graph shows the estimated coefficients $\{\beta\}$ as well as 95% confidence intervals obtained from standard errors clustered at the PIN Code level.

coefficient β for the specification in Eq. (1), using these portfolio shares as the dependent variable in place of brown share. The results are presented in Panels A, B, and C of Appendix Table A10, respectively. We do not observe any shifts in the sensitivity of these other portfolio characteristics to pollution after the introduction of CAAQMSs.²⁶

Investor Preferences, Beliefs, and the Consequences of Pollution Information for Returns: The question naturally arises of whether portfolio adjustments are made as a result of beliefs about the returns of green versus brown stocks, or a distaste for holding brown stocks. To explore this distinction, we begin by showing in Appendix Figure A6 the cumulative returns of brown and green portfolios over the period January 2000 to December 2019 (see Section 3 for details on the portfolios' construction). Over any relevant horizon, the two portfolios perform quite similarly, though with the brown portfolio actually generating higher returns. On the face of it, these broad patterns suggest that, if investors sold brown stocks because of beliefs of low returns, they were mistaken on average.

To examine the consequences of selling brown stock sales that can be tied specifically to sales resulting from pollution information, we conducted a counterfactual analysis to assess whether selling induced by air pollution information on high-pollution dates is costly for investors in terms of future returns. Specifically, we estimate the counterfactual returns that investors would have earned from the brown

stocks they sold on high-pollution days (defined as those with pollution levels above the median of distribution for all PIN-Code-days), had they held onto these stocks for some additional period of time (we consider windows of 1, 3, and 6 months) beyond the actual selling date. This estimation is conducted before and after the arrival of a monitoring station to capture the impact of increased pollution information. Our estimating equation is as follows:

$$Return_{p(m),t}^{High\ Pollution} = \beta \times Treat_p \times Post_{m,t} + X'_{p,t} \theta + \gamma_p + \lambda_{m,t} + \varepsilon_{p,t} \quad (6)$$

where $Return_{p(m),t}^{High\ Pollution}$ refers to the counterfactual return that could be generated by the brown stocks sold by an average investor in PIN Code p on high pollution days during month t , over a hypothetical one-, three-, or six-month period immediately after the actual selling date. The parameter β captures changes before and after the arrival of a monitoring station. We further augment the preceding analysis by looking at returns sold on high- versus low-pollution days (pollution levels above/below the median of the sample distribution), by using as the dependent variable the difference, $Return_{p(m),t}^{High\ Pollution} - Return_{p(m),t}^{Low\ Pollution}$, which captures the difference in the hypothetical one-, three-, six-month return that investors could have earned from the brown stocks they sold on high versus low pollution days.

The results are detailed in Appendix Table A11 for counterfactual returns on high pollution days (Panel A) and high-versus-low pollution days (Panel B). We find that brown stocks sold on high-pollution days generate statistically significant positive excess returns after pollution information becomes available, relative to the pre-period. For instance, $\beta = 0.1246$ in Column 4 of Panel A corresponds to an annualized counterfactual market-adjusted return of approximately 1.5% (i.e., 0.1246×12). While it is beyond the scope of this paper to fully understand the reasons behind these positive counterfactual returns, our results indicate a real economic cost in terms of foregone trading profits from pollution-induced trading. When analyzing holding periods of three and six months, we continue to find that holding onto brown stocks—particularly after the arrival of a monitoring station—would lead to higher profits. However, these estimates are less precisely estimated and not statistically significant. To the extent that investors anticipate these lower returns, our results align with the notion that investors derive a warm-glow utility from green investments.

5.4. Response to high-salience changes in air quality

Thus far, we have posited that the increased salience of pollution via CAAQMSs is responsible for the increased sensitivity of investors' green portfolios to air quality. In this section, we explore a potentially more direct link to salience as an explanation, based on high-frequency changes around particularly salient shifts in air quality. To do so, we look at transitions across the color-coded Air Quality Index (AQI) categories used by the Central Pollution Control Board to communicate pollution severity to the public. We focus on the transition from Moderate (yellow) to Poor (amber) at an AQI cutoff of 200, which is particularly important in terms of government communication. The government issues “amber alert” messages via social media and elsewhere on days when the AQI exceeds 200, warning that prolonged exposure to outside air may cause breathing discomfort and respiratory problems; the government also recommends the use of masks when the AQI is above 200.²⁷

²⁷ Government messaging continues at higher AQIs, with more dire messages as the AQI transitions to red (“respiratory illness with long exposure”). We do not emphasize this transition in the main text for two reasons. First, we see a sharper distinction between yellow and amber in the sense that it causes a shift from no-warning to warning, as compared to simply a transition to a marginally more extreme warning. Second, pollution in the high-200s and above is arguably so severe that pollution becomes very salient irrespective of any color coding. We do provide results on the amber-to-red transition

²⁶ In unreported results, we confirm that there is little shift in the average stock characteristics for the portfolio held by local investors.

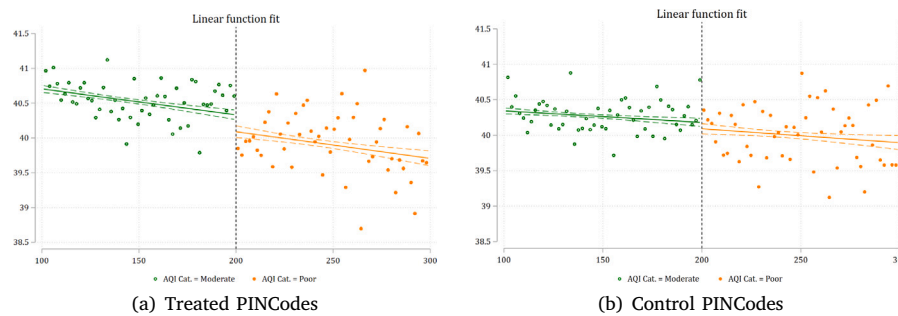


Fig. 8. Regression discontinuity plots for brown share as a function of air quality readings.

Note: The figure presents RD plots for PIN-Code-day-level portfolio brown shares among treated (left panel) and control (right panel) investors as a function of local AQI readings. We plot the mean value of portfolio brown shares, binned by AQI, as well as the fitted local linear trend (along with 95% confidence intervals) around the critical AQI value of 200, which is the cutoff between “Moderate” (yellow) and “Poor” (amber) pollution.

Table 3

Regression discontinuity estimates: AQI transition from “Moderate” to “Poor”.

(1)	(2)	(3)	(4)	(5)	(6)
smoothness constant	M	band-width	estimate	Lower CI	Upper CI
Panel A Treated PIN Codes					
Kernel = Uniform					
0.1*M _{rot}	0.0001	41	−0.65	−0.8297	−0.4703
0.5*M _{rot}	0.0007	21	−0.9145	−1.1638	−0.6652
M _{rot}	0.0014	16	−0.8878	−1.1747	−0.6009
5*M _{rot}	0.0071	8	−0.6721	−1.0825	−0.2616
10*M _{rot}	0.0143	6	−0.9334	−1.4149	−0.4519
Kernel = Triangle					
0.1*M _{rot}	0.0001	52.36	−0.6813	−0.8565	−0.5061
0.5*M _{rot}	0.0007	27.85	−0.8361	−1.0793	−0.593
M _{rot}	0.0014	20.92	−0.8513	−1.1321	−0.5705
5*M _{rot}	0.0071	11.20	−0.7361	−1.1386	−0.3337
10*M _{rot}	0.0143	8.38	−0.7461	−1.2204	−0.2718
Panel B Control PIN Codes					
Kernel = Uniform					
0.1*M _{rot}	0.0001	40	−0.117	−0.2712	0.0372
0.5*M _{rot}	0.0007	21	−0.1156	−0.3298	0.0986
M _{rot}	0.0013	15	−0.0892	−0.3372	0.1588
5*M _{rot}	0.0067	8	−0.0001	−0.3511	0.3509
10*M _{rot}	0.0134	6	−0.167	−0.5776	0.2436
Kernel = Triangle					
0.1*M _{rot}	0.0001	50.64	−0.1372	−0.2874	0.013
0.5*M _{rot}	0.0007	26.94	−0.0812	−0.2898	0.1275
M _{rot}	0.0013	20.24	−0.0999	−0.3412	0.1413
5*M _{rot}	0.0067	10.83	−0.1169	−0.4601	0.2263
10*M _{rot}	0.0134	8.16	−0.1439	−0.5453	0.2575

Note: The values reported here are RD estimates around the AQI cutoff of 200 at which the air quality index category switches from “Moderate”(yellow) to “Poor”(amber). Panels A and B presents the estimates for the treated and control PIN Codes, respectively. We follow the inference procedure of [Armstrong and Kolesár \(2018\)](#) to select the optimal bandwidth and to construct the confidence interval. M_{rot} indicates the rule-of-thumb choice of the smoothness constant, M. We also vary M as form of sensitivity analysis. The unit of observation is PIN-Code-day.

We study responsiveness around the yellow-amber transition using PIN-Code-day observations in the range of 100 to 300 (i.e., 100 units above and below the cutoff). We begin by presenting visual evidence of this relationship in the left panel of [Fig. 8](#), which provides a regression

in Appendix Figure A7, and do not find any significant shift around this change. Finally, note that the Central Pollution Control Board of India uses six categories in total, with each level accompanied by different health advisories. AQI statistics and color categories are available at https://airquality.cpcb.gov.in/AQI_All_India/, last accessed April 23, 2025.

discontinuity plot for local brown share, using distance to the “yellow-amber” threshold of 200 as the running variable and a local linear trend on each side of the discontinuity. Consistent with our previous findings, we observe a negative relation between local air quality and brown share, at the higher frequency of observation we use in this analysis. In addition, we observe a clear discontinuity at the yellow-amber cutoff, with a discernable drop in portfolio brown share at AQI = 200 (i.e., the level that triggers amber alert warnings). In the right panel of [Fig. 8](#), we show the analogous pattern for control Pin Codes and observe no such discontinuous effect (nor do we see a general negative relationship between pollution and brown share).

We provide the results of our RD analysis in [Table 3](#), following the procedure outlined by [Armstrong and Kolesár \(2018\)](#) to select the optimal bandwidth and construct confidence intervals. Specifically, we begin with the rule-of-thumb choice of the smoothness constant, M_{rot} , and also confirm robustness using different choices. Across all specifications in [Table 3](#), the coefficient of interest is negative and statistically significant. In economic terms, Panel A shows that portfolio brown share is 0.65 to 0.91 percentage points (1.66% to 2.22% in relative terms) lower when AQI is just above the amber alert cutoff.²⁸

Overall, we take our RD estimates as bolstering the salience-based interpretation of our main results, given the pronounced shift in brown share portfolio allocations above a threshold that triggers greater visibility of the harmfulness of pollution.

5.5. Heterogeneity by investor type

We provide some exploratory analyses based on several attributes of traders that potentially relate to propensity to access local pollution information and/or concern for the environment, including age, gender, and whether investors primarily execute trades via mobile, internet, or more traditional means.

We naturally do not have random assignment or any close approximation to it for these attributes, and as such the results should be interpreted with caution. Still, the patterns are interesting to consider as an extension to our main results given that, based on intuition as well as past research, one may have prior expectations about which groups will have greater sensitivity to pollution salience.

Tech-Savvy Investors: We begin by comparing sensitivity of brown share to pollution based on whether an investor predominantly makes trades via (a) mobile, (b) internet, or (c) broker or other “traditional” method. As observed in the introduction, given that the public predominantly obtained AQI updates via smartphone, it is natural to speculate

²⁸ As already suggested by our graphical results, we do not find any significant effect when AQI transitions from amber to red, which we show in Appendix Table A12.

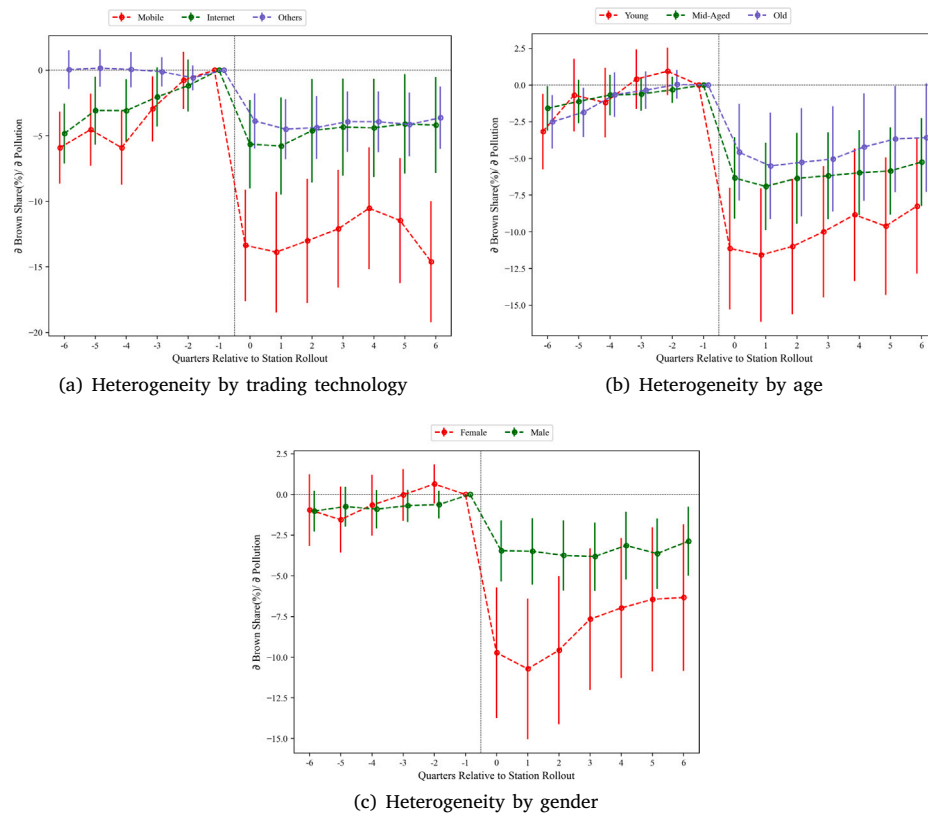


Fig. 9. Heterogeneity analyses of changes in brown-share-pollution gradient.

Note: This figure shows the sensitivity of brown investment to pollution before and after the installation of Continuous Ambient Air Quality Monitoring Stations for different groups of investors. Specifically we plot the coefficients $\{\beta_i\}$ from our preferred specification in Table 2, column 8, splitting the investor sample by various attributes. Panel A shows heterogeneity by trading technology, Panel B shows heterogeneity by age, and Panel C shows heterogeneity by gender. In each case, the graph shows the estimated coefficients $\{\beta_i\}$ as well as 95% confidence intervals obtained from standard errors clustered at the PIN Code level.

that more tech-savvy investors — who use their mobile devices to execute trades — would be more exposed to pollution updates.

In Fig. 9, Panel A (the corresponding regression results are in Appendix Table A13), we show event study plots for responsiveness to air quality information disaggregated by investors' primary means of trading. As may readily be seen in the graph, the largest effect is observed among investors who use mobile devices to trade. The confidence intervals of the other two groups — internet and physical trading — are largely overlapping, though the effect of monitoring stations is marginally greater for internet-based traders.

Investors' Demographic Profiles: We next turn to explore heterogeneity by age. Before presenting these results, it is worth noting that while younger investors are more frequently classified as “mobile” traders, the differences are surprisingly modest relative to middle-aged investors: 13 percent of younger investors are classified as mobile, as compared to 10 percent of middle-aged investors (elderly investors are far less likely to use mobile devices for trades—only 3.7 percent of the elderly in our sample are classified as mobile investors).

More importantly, putting aside technological concerns, there are reasons to expect differential responsiveness to pollution information by age. First, a vast literature documents a very strong negative correlation between environmental concerns and age—unsurprising, given that the young will disproportionately bear the consequences of climate change and environmental degradation.²⁹ Of more direct relevance,

recent survey-based evidence finds higher stated interest in ESG among younger investors and lower interest among older investors (Haber et al., 2022). An alternative hypothesis is that the elderly might be more sensitive to information on air quality, because they are far more vulnerable to the effects of air pollution (see Gouveia and Fletcher, 2000; Fischer et al., 2003).

In Fig. 9, Panel B, we revisit our main event plot, disaggregating the sample into young (18–29), middle-aged (30–55), and elderly (above 56) investors; the corresponding regression results are in Appendix Table A14. We observe a substantially greater shift in brown-investment-to-pollution sensitivity among the young, relative to the other two age groups. While we cannot put too strong an interpretation around this finding, we see this particular result as reinforcing the above-cited evidence on the greater interest in ESG investing for younger investors.

Finally, we split the sample by gender. As noted in the introduction, there is an *ex ante* rationale for a differential response given past work on a “gender environmentalism gap” (e.g., Xiao and McCright, 2015). We show the differential response by gender in the event plots in Fig. 9 Panel C. The gender difference is striking—women exhibit a responsiveness that is 2–3 times greater than that of men. Observe that the male point estimates are more precisely estimated and also much closer to the full-sample estimates, which reflects the fact that most Indian retail investors are men. For completeness, we also provide the tabular version of the gender split in Appendix Table A15, which provides the same message as the event plots.

²⁹ See, e.g., Liere and Dunlap (1980), for an early and well-cited review which describes age as the “predominant” individual attribute that is correlated with environmental concerns; a more recent review article

by Sanchez-Sabate and Sabaté (2019) similarly finds an important role for age, focused specifically on environmental concerns and meat consumption.

Our final “heterogeneity” test explores whether we see a comparable shift in responsiveness for institutional investors as is observed in our sample of retail investors, which has been our focus to this point. We present results on institutional investors’ trades in Appendix Figure A8. If anything, institutional investors’ response to the availability of air quality information moves in the opposite direction to that of retail investors, though we cannot reject that institutions are simply unaffected by the appearance of real-time pollution data. There are various potential explanations for this non-result. The most natural is that institutional investors may be less sensitive to “taste-based” shifts in investing. However, we may also simply have a less-precise mapping of AQI to relevant location, since the PIN Code for an institutional investor reflects their place of work rather than their residence. More broadly, we interpret this non-result with caution, relative to our main findings on retail investors.³⁰

6. Conclusion

We document that exposing investors to ready information about air pollution heightens the sensitivity of their “brown” investments to air quality. We interpret these findings through the lens of salience, in the spirit of [Bordalo et al. \(2013\)](#) among others—ready access to air quality data makes this information a more salient input into investment decisions.

As we noted in our discussion of the results, this shift comes despite the fact that investors may reduce their returns as a result of these pollution-induced trades. That said, we cannot identify whether the shift we document is driven by mistaken beliefs, or shifts in investor tastes as a result of greater attentiveness to environmental problems. Distinguishing between these two explanations is one important future direction, and one that we plan to pursue going forward.

Finally, our findings highlight a role for salience in driving portfolio allocations. To the extent that it is desirable to shift investments away from brown firms (itself a debated point), it may be useful to implement policies that highlight environmental problems in ways that draw investors’ attention to such concerns.

CRediT authorship contribution statement

Raymond Fisman: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pulak Ghosh:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Arkodipta Sarkar:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jian Zhang:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

³⁰ We also considered heterogeneity based on whether companies were headquartered in the same district as the investor (an investor-firm pair attribute). Specifically, we construct separate local and non-local brown portfolio shares based on this same-district definition, and compare the impact of the appearance of a CAAQMS on the sensitivity of each to local air pollution. We find a larger impact of pollution information on the non-local brown share. Since there are numerous candidate explanations for this finding, we do not emphasize it, though it does suggest that our main results are not driven entirely by investors’ immediate concern for their own health and well-being.

Acknowledgment

Jian Zhang acknowledges financial support from the Research Grant Council of Hong Kong (Grant No. 17504925) and University of Hong Kong Seed Fund for Basic Research (Grant No. 109001081).

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jfineco.2026.104309>.

Data availability

Replication package is available at <https://data.mendeley.com/datasets/kwjwcn3wgy/2>.

References

- Agee, M.D., Atkinson, S.E., Crocker, T.D., Williams, J.W., 2014. Non-separable pollution control: Implications for a CO2 emissions cap and trade system. *Resour. Energy Econ.* 36 (1), 64–82.
- Akey, P., Appel, I., 2019. Environmental externalities of activism. Available At SSRN 3508808.
- An, L., Huang, S., Lou, D., Wen, X., Xu, M., 2024. Inferring Mutual Fund Intra-Quarter Trading: An Application to ESG Window Dressing. Working paper.
- Andersen, S., Chebarov, D., Filali Adib, F.Z., Meisner Nielsen, K., 2024. A breath of change: Can personal exposures drive green preferences? Indiana University working paper.
- Armstrong, T.B., Kolesár, M., 2018. Optimal inference in a class of regression models. *Econometrica* 86 (2), 655–683.
- Baker, A.C., Larcker, D.F., Wang, C.C., 2022. How much should we trust staggered difference-in-differences estimates? *J. Financ. Econ.* 144 (2), 370–395.
- Balasubramaniam, V., Campbell, J.Y., Ramadorai, T., Ranish, B., 2023. Who owns what? A factor model for direct stockholding. *J. Financ.* 78 (3), 1545–1591.
- Barwick, P.J., Li, S., Lin, L., Zou, E., 2019. From Fog to Smog: The Value of Pollution Information. Technical Report, National Bureau of Economic Research.
- Barwick, P.J., Li, S., Lin, L., Zou, E.Y., 2024. From fog to smog: The value of pollution information. *Am. Econ. Rev.* 114 (5), 1338–1381.
- Bau, N., Matray, A., 2023. Misallocation and capital market integration: Evidence from India. *Econometrica* 91 (1), 67–106.
- Bauer, R., Ruof, T., Smeets, P., 2021. Get real! Individuals prefer more sustainable investments. *Rev. Financ. Stud.* 34 (8), 3976–4043.
- Ben-David, I., Hirshleifer, D., 2012. Are investors really reluctant to realize their losses? Trading responses to past returns and the disposition effect. *Rev. Financ. Stud.* 25 (8), 2485–2532.
- Berg, F., Fabisik, K., Sautner, Z., 2021. Is history repeating itself? The (un) predictable past of ESG ratings. *SSRN Electron. J.* 1–59.
- Berg, F., Koelbel, J.F., Pavlova, A., Rigobon, R., 2022a. ESG Confusion and Stock Returns: Tackling the Problem of Noise. Technical Report, National Bureau of Economic Research.
- Berg, F., Koelbel, J.F., Rigobon, R., 2022b. Aggregate confusion: The divergence of ESG ratings. *Rev. Financ.* 26 (6), 1315–1344.
- Berry, S.T., 1994. Estimating discrete-choice models of product differentiation. *Rand J. Econ.* 242–262.
- Boehmer, E., Jones, C.M., Zhang, X., Zhang, X., 2021. Tracking retail investor activity. *J. Financ.* 76 (5), 2249–2305.
- Bolton, P., Kacperczyk, M., 2021. Do investors care about carbon risk? *J. Financ. Econ.* 142 (2), 517–549.
- Bordalo, P., Gennaioli, N., Shleifer, A., 2013. Salience and asset prices. *Am. Econ. Rev.* 103 (3), 623–628.
- Callaway, B., Sant’Anna, P.H., 2021. Difference-in-differences with multiple time periods. *J. Econometrics* 225 (2), 200–230.
- Cengiz, D., Dube, A., Lindner, A., Zipperer, B., 2019. The effect of minimum wages on low-wage jobs. *Q. J. Econ.* 134 (3), 1405–1454.
- Chang, T.Y., Huang, W., Wang, Y., 2018. Something in the air: Pollution and the demand for health insurance. *Rev. Econ. Stud.* 85 (3), 1609–1634.
- Choi, D., Gao, Z., Jiang, W., 2020. Attention to global warming. *Rev. Financ. Stud.* 33 (3), 1112–1145.
- Cosemans, M., Frehen, R., 2021. Salience theory and stock prices: Empirical evidence. *J. Financ. Econ.* 140 (2), 460–483.
- De Janvry, A., He, G., Sadoulet, E., Wang, S., Zhang, Q., 2023. Subjective performance evaluation, influence activities, and bureaucratic work behavior: Evidence from China. *Am. Econ. Rev.* 113 (3), 766–799.
- Fischer, P., Hoek, G., Brunekreef, B., Verhoef, A., Van Wijnen, J., 2003. Air pollution and mortality in The Netherlands: are the elderly more at risk? *Eur. Respir. J.* 21 (40 suppl), 34s–38s.

- Frydman, C., Hartzmark, S.M., Solomon, D.H., 2018. Rolling mental accounts. *Rev. Financ. Stud.* 31 (1), 362–397.
- Galiani, S., Gertler, P., Schargrodsky, E., 2005. Water for life: The impact of the privatization of water services on child mortality. *J. Political Econ.* 113 (1), 83–120.
- Giglio, S., Maggiori, M., Stroebe, J., Tan, Z., Utkus, S., Xu, X., 2023. Four Facts About ESG Beliefs and Investor Portfolios. Technical Report, National Bureau of Economic Research.
- Goldberg, P.K., Khandelwal, A.K., Pavcnik, N., Topalova, P., 2010. Imported intermediate inputs and domestic product growth: Evidence from India. *Q. J. Econ.* 125 (4), 1727–1767.
- Goldstein, I., Kopytov, A., Shen, L., Xiang, H., 2022. On ESG Investing: Heterogeneous Preferences, Information, and Asset Prices. Technical Report, National Bureau of Economic Research.
- Goodman-Bacon, A., 2021. Difference-in-differences with variation in treatment timing. *J. Econometrics* 225 (2), 254–277.
- Gouveia, N., Fletcher, T., 2000. Time series analysis of air pollution and mortality: effects by cause, age and socioeconomic status. *J. Epidemiology & Community Health* 54 (10), 750–755.
- Graff Zivin, J., Neidell, M., 2013. Environment, health, and human capital. *J. Econ. Lit.* 51 (3), 689–730.
- Greenstone, M., He, G., Jia, R., Liu, T., 2022. Can technology solve the principal-agent problem? Evidence from China's war on air pollution. *Am. Econ. Rev.: Insights* 4 (1), 54–70.
- Gulia, S., Shukla, N., Padhi, L., Bosu, P., Goyal, S., Kumar, R., 2022. Evolution of air pollution management policies and related research in India. *Environ. Challenges* 6, 100431.
- Haber, S., Kepler, J.D., Larcker, D.F., Seru, A., Tayan, B., 2022. ESG investing: What shareholders do fund managers represent? Rock Center for Corporate Governance at Stanford University Working Paper.
- Hartzmark, S.M., Shue, K., 2023. Counterproductive impact investing: The impact elasticity of brown and green firms. Available At SSRN 4359282.
- Hartzmark, S.M., Sussman, A.B., 2019. Do investors value sustainability? A natural experiment examining ranking and fund flows. *J. Financ.* 74 (6), 2789–2837.
- Heath, D., Macciocchi, D., Michaely, R., C. Ringgenberg, M., 2023. Does socially responsible investing change firm behavior? *Rev. Financ.* 27 (6), 2057–2083.
- Huberman, G., Regev, T., 2001. Contagious speculation and a cure for cancer: A nonevent that made stock prices soar. *J. Financ.* 56 (1), 387–396.
- Kahn, M.E., Matsusaka, J., Shu, C., 2023. Divestment and Engagement: The Effect of Green Investors on Corporate Carbon Emissions. Technical Report, National Bureau of Economic Research.
- Kang, L., Jiang, H., Nie, Z.Z., Zhou, H., 2024. Can old sin make new shame? Stock market reactions to the release of movies re-exposing past corporate scandals. *Financ. Res. Lett.* 105755.
- Kaniel, R., Saar, G., Titman, S., 2008. Individual investor trading and stock returns. *J. Financ.* 63 (1), 273–310.
- Khanna, T., Palepu, K., 2000. Is group affiliation profitable in emerging markets? An analysis of diversified Indian business groups. *J. Financ.* 55 (2), 867–891.
- Koijen, R.S., Richmond, R.J., Yogo, M., 2023. Which investors matter for equity valuations and expected returns? *Rev. Econ. Stud.* rdad083.
- Krey, V., Masera, O., Blanford, G., Bruckner, T., Cooke, R., Fisher-Vanden, K., Haberl, H., Hertwich, E., Kriegler, E., Mueller, D., et al., 2014. Annex 2-metrics and methodology.
- Krueger, P., Sautner, Z., Starks, L.T., 2020. The importance of climate risks for institutional investors. *Rev. Financ. Stud.* 33 (3), 1067–1111.
- Li, Y., Johnson, E.J., Zaval, L., 2011. Local warming: Daily temperature change influences belief in global warming. *Psychol. Sci.* 22 (4), 454–459.
- Li, Q., Watts, E.M., Zhu, C., 2023. Retail investors and ESG news. Jacobs Levy Equity Management Center for Quantitative Financial Research Paper.
- Liere, K.D.V., Dunlap, R.E., 1980. The social bases of environmental concern: A review of hypotheses, explanations and empirical evidence. *Public Opin. Q.* 44 (2), 181–197.
- Lin, X., Yang, R., Zhang, W., Zeng, N., Zhao, Y., Wang, G., Li, T., Cai, Q., 2023. An integrated view of correlated emissions of greenhouse gases and air pollutants in China. *Carbon Balance Manag.* 18 (1), 9.
- Lujala, P., Lein, H., Rød, J.K., 2015. Climate change, natural hazards, and risk perception: the role of proximity and personal experience. *Local Environ.* 20 (4), 489–509.
- Moss, A., Naughton, J.P., Wang, C., 2023. The irrelevance of environmental, social, and governance disclosure to retail investors. *Manag. Sci.*
- Naaraayanan, S.L., Sachdeva, K., Sharma, V., 2021. The real effects of environmental activist investing. *Eur. Corp. Gov. Institute-Finance Work. Pap.* (743).
- Oehmke, M., Opp, M.M., 2024. A theory of socially responsible investment. *Rev. Econ. Stud.* rdae048.
- Pant, P., Lal, R.M., Guttikunda, S.K., Russell, A.G., Nagpure, A.S., Ramaswami, A., Peltier, R.E., 2019. Monitoring particulate matter in India: recent trends and future outlook. *Air Qual. Atmosphere & Health* 12 (1), 45–58.
- Pástor, L., Stambaugh, R.F., Taylor, L.A., 2021. Sustainable investing in equilibrium. *J. Financ. Econ.* 142 (2), 550–571.
- Pastor, L., Stambaugh, R.F., Taylor, L.A., 2023. Green Tilts. Technical Report, National Bureau of Economic Research.
- Pedersen, L.H., Fitzgibbons, S., Pomorski, L., 2021. Responsible investing: The ESG-efficient frontier. *J. Financ. Econ.* 142 (2), 572–597.
- Pelizzon, L., Rzezniak, A., Hanley, K.W., 2021. The salience of ESG ratings for stock pricing: Evidence from (potentially) confused investors.
- Rees, N., Wickham, A., Chandy, L., 2019. Silent suffocation in Africa. UNICEF Rep..
- Roychowdhury, A., Somvanshi, A., Kaur, S., 2023. Status of air Quality Monitoring in India: Spatial Spread, Population Coverage and Data Completeness. Urban Lab - Centre for Science and Environment Analysis.
- Sanchez-Sabate, R., Sabaté, J., 2019. Consumer attitudes towards environmental concerns of meat consumption: A systematic review. *Int. J. Environ. Res. Public Health* 16 (7), 1220.
- Starks, L.T., Venkat, P., Zhu, Q., 2023. Corporate ESG profiles and investor horizons. Available At SSRN 4312426.
- Sun, L., Abraham, S., 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *J. Econometrics* 225 (2), 175–199.
- Tsai, T.-C., Jeng, Y.-J., Chu, D.A., Chen, J.-P., Chang, S.-C., 2011. Analysis of the relationship between MODIS aerosol optical depth and particulate matter from 2006 to 2008. *Atmos. Environ.* 45 (27), 4777–4788.
- Van Donkelaar, A., Martin, R.V., Park, R.J., 2006. Estimating ground-level PM_{2.5} using aerosol optical depth determined from satellite remote sensing. *J. Geophys. Res.: Atmospheres* 111 (D21).
- Xiao, C., McCright, A.M., 2015. Gender differences in environmental concern: Revisiting the institutional trust hypothesis in the USA. *Environ. Behav.* 47 (1), 17–37.
- Zaval, L., Keenan, E.A., Johnson, E.J., Weber, E.U., 2014. How warm days increase belief in global warming. *Nat. Clim. Chang.* 4 (2), 143–147.